



About

Real Options for Adoption and Resilience:

Tutorials

Description: Analytical, Finite Difference, Maximum Likelihood and Monte Carlo methods—a complete set of functions for maximum likelihood estimation and the calculation of probabilities, option prices, decision thresholds, visiting times, first passage times and more—everything for a real options analysis.

Code: Excel user-defined functions, Visual Basic for Applications code, R6 objects and an RShiny App.

Help: Shiny Tutorials—step-by-step instructions for the Shiny App.
Reference—explanations for all functions in Excel and R6.

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Versions: Excel: 1.4.5.0 (Stochastic Process.Code.Help.Bugs)
R: 1.3.5.0 (Stochastic Process.Code.Help.Bugs)

Citation:

Hertzler, Greg, Capon, Tim. (2026). *Real Options for Adoption and Resilience: Tutorials*. Publisher. https://greghertzler.github.io/GregsOUPShiny/OUP_Shiny.html

Using Help:

There are three panes: a Navigate Pane on the left, a Find Pane in the upper left and a Topics Pane on the right. Between the panes, click '<<' to close and '>>' to open the Navigate and Find Panes.

On the Navigate Pane, click the tabs to jump to topics. Clicking a tab will expand or collapse its subsections. Click '≡' to expand all subsections. Click '-' to collapse all subsections.

At the upper right and lower right of the Topics Pane, click '↑' for the previous topic and '↓' for the next topic. Click the browser's back and forward arrows to revisit topics.

On the Find Pane, type keywords into the box and click '>>' or press 'Enter'. Tabs in the Navigate Pane will be marked to identify which topics contain the keywords. Paragraphs containing the keywords will be marked in the Topics Pane. To clear the marks, click '<<' or click the '×' in the box and press 'Enter'.



Tutorials

What brings you here?

Perhaps you have a problem to solve or a research question to answer. These tutorials are organised into Case Studies and Methods. The Case Studies emphasise questions and answers. The Methods demonstrate calculations to transform questions into answers.

The guide below poses questions and provides links to relevant Case Studies and Methods.

Questions

- Resilience
 - How quickly will a system recover after shocks?
Case Study: Wandering Albatross
Method: Recovery
 - When will a system change irreversibly?
Case Study: Farm Dams and Destocking
Method: Tipping Point
- Adoption
 - How will agriculture adapt to climate change?
Case Study: Agriculture and Climate
Method: Entry Option
Exit Option
- Policies
 - Will charging resource rent discourage industry?
Case Study: Fisheries and ITQs
Method: Entry Option
Resource Rent
 - Why are producers reluctant to sign contracts?
Case Study: Carbon Contracts
Method: Contracts to Deliver
 - Is Yield Index Insurance a viable drought policy?
Case Study: Yield Index Insurance
Method: Other Options
- Finance
 - How does finance with the Ornstein-Uhlenbeck Process differ from finance with Brownian Motion?
Method: Vanilla Call Option
Vanilla Put Option
Bear Spread

- How can I price vanilla options, exotic options, derivatives and insurance?
 - Method: Vanilla Call Option
 - Vanilla Put Option
 - Strangle and Straddle
 - Bear Spread
 - Weather Derivative
 - Other Options
- When is my stop loss order likely to be triggered?
 - Method: Time to Stop Loss
- Estimation
 - How do I estimate the Ornstein-Uhlenbeck Process?
 - Method: Rate of Convergence
 - Observation Interval
 - How do I interpret the parameters?
 - Method: Rate of Convergence
 - Observation Interval
 - Smoothed Data
 - Sample Size
 - Do I fill in for missing observations or not?
 - Method: No Missing Observations
 - Must I use raw data or can I use smoothed data?
 - Method: Smoothed Data
 - How many observations are enough?
 - Method: Sample Size
 - How do I test hypotheses about the parameters?
 - Method: Hypothesis Tests
 - How do I organise my data?
 - Method: My Data

Or, perhaps you are interested in a particular method, for example, option pricing. You can find appropriate tutorials by typing 'option' in the Find box in the upper left of the screen and clicking the '>>' button. All the tutorials using 'option' will be highlighted for you to browse. If 'option' is boring, find something more interesting like 'passage time'. When you are done, click the '<<' button to clear.

Or, perhaps your interest is more general. Under the Ornstein-Uhlenbeck menu, the Real Options panel organises the calculations for a Real Options analysis. Also under the menu, the Analytical, Finite Difference and Maximum Likelihood panels calculate functions in order of increasing difficulty. You can step through the tabs on each of these panels to see what's possible. Click the Info buttons to read short explanations. Then you might browse the tutorials to see what's interesting. If you are still keen, under the Help menu, you will find the Reference.

Case Studies

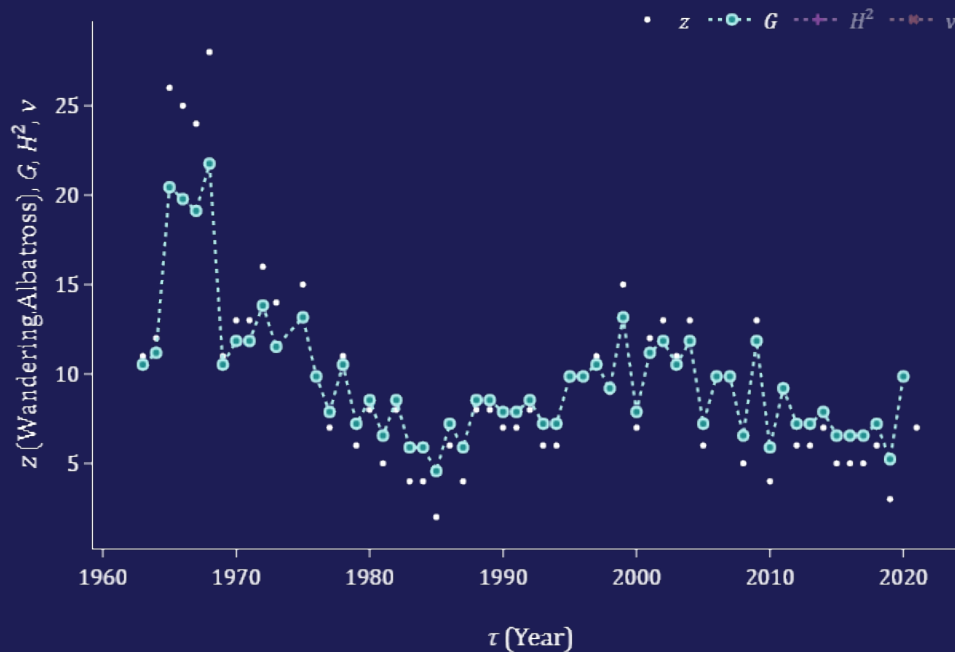
Case Studies use the methods of Real Options to answer research questions. Here we emphasise the questions and answers. At the bottom of each case study are links to the relevant methods.

Wandering Albatross

Depending upon whom you ask, Wandering Albatross are classified as Vulnerable or Critically Endangered. Danger-defying scientists in Tasmania have been counting the eggs in the nesting sites since 1963. The egg counts are publicly available.¹

How quickly can the Wandering Albatross recover from a bad egg year?

On the plot below, egg counts, z , are on the vertical axis and years, τ , are on the horizontal axis. An observation includes four numbers: the current year, the current egg count, the next year, and the next egg count. Each observation has its own Mean, G . A Mean is the expected value of next year's egg count, given the current year and the current egg count. Each observation also has its own Variance, H^2 , which can be, but in this case is not, shown on the plot.



The Ornstein-Uhlenbeck parameters for the Albatross egg counts are in this table.

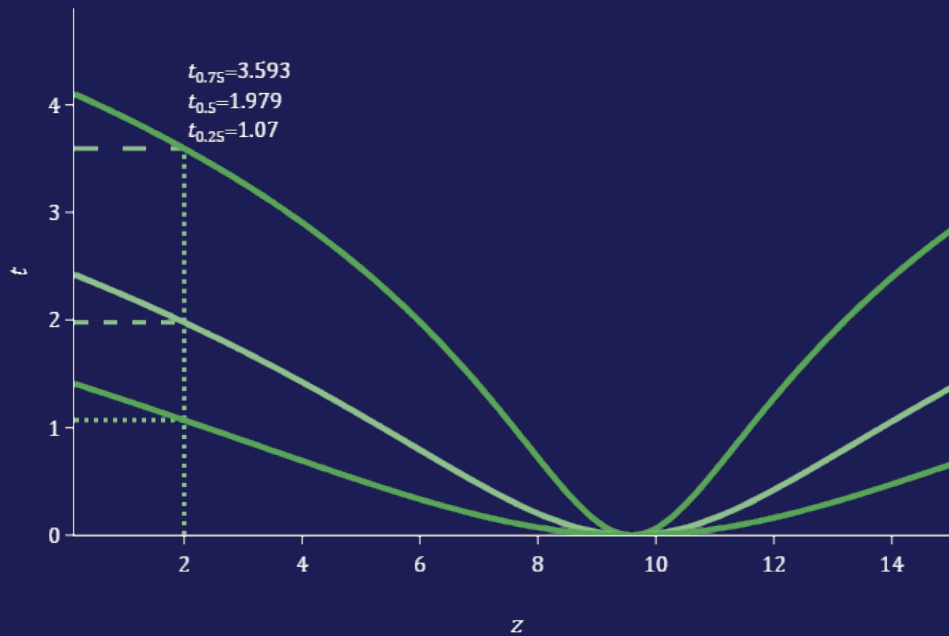
	rho	mu	sigma	alpha
Parameters	0.413631	9.59138	5.03194	0.716464

Parameter rho is the rate of convergence; parameter mu is the location and parameter sigma is the scale. Given enough time, a Mean will converge to the long-term mean, mu. Given half as much time, a Variance will converge to the long-term variance, $\sigma^2/(2\rho)$. Parameter alpha is determined by the rate of convergence and the length of time for an observation. It summarises how the Means and Variances converge.

If alpha is 0.5, the process is stationary or ergodic and the Means and Variances have converged every time eggs are counted. If alpha is 1, the process is a random walk, or Brownian Motion, and the Means and Variances never converge. Egg counts are about half-way between stationary and a random walk. After a shock, egg counts will converge toward the long-term mean, but not quickly.

¹The Australian Threatened Species Index, <https://tsx.org.au/tsx2024>

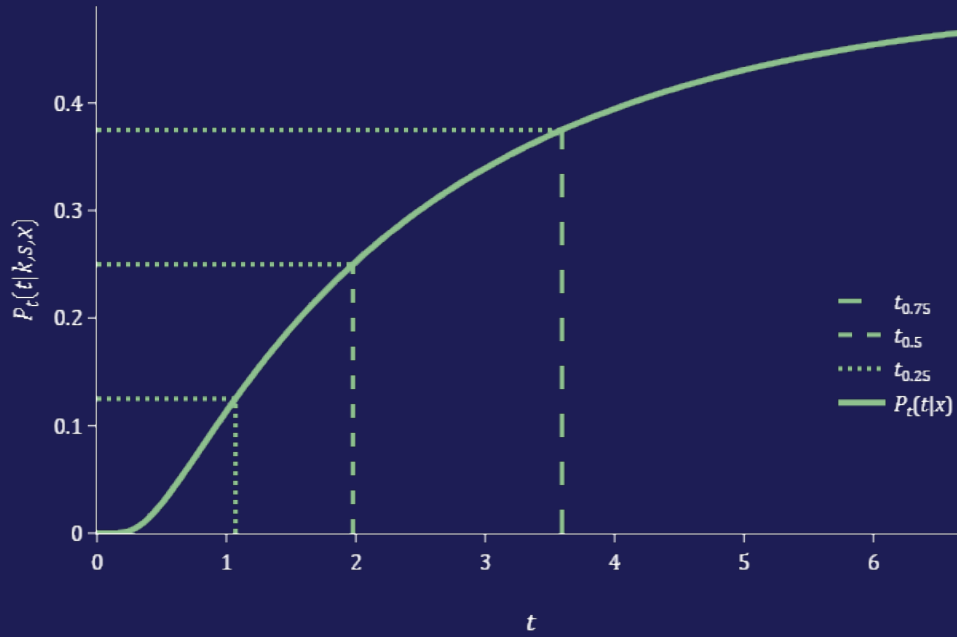
The plot below shows times to recover to the long-term mean from all possible egg counts. Time is on the vertical axis and egg counts are on the horizontal axis. The curves are for percentiles $t_{0.25}$, $t_{0.5}$ and $t_{0.75}$. The worst egg year, with only two eggs, is an example.



Because the long-term mean is not a tipping point, recovery is reversible and times are Visiting Times. Starting from two eggs, there is a 25% chance of recovering before just over a year has passed, a 50% chance of recovering before just short of two years have passed and a 75% chance of recovering before just over 3½ years have passed.

This seems like good news. But what does recovery actually mean?

The plot below shows all possible Visiting Times on the horizontal axis and Visiting Time Probabilities on the vertical axis.



Visiting Times are doubly confusing. Notice the probability does not go to one. A Visiting Time Probability is the proportion of time spent above a reversible threshold. If the reversible threshold is the long-term mean, the Visiting Time Probability reaches a maximum of 0.5. Once egg counts reach the long-term mean, they are expected to be above the long-term mean 50% of the time and below the long-term mean 50% of the time. The median Visiting Time is when there is a 50% chance the egg counts will be above the long-term mean 50% of the time. On the plot, this is the time when the probability equals 0.25.

Of course there are caveats. As many optimists have said, 'Every model is wrong but some are useful.' Or as a pessimist might say, 'Every model is a collection of unrealistic assumptions we are willing to make because we are desperate to calculate an answer. If we are being honest, we really don't know if our answer is useful.'

So how desperate are we? The Ornstein-Uhlenbeck Process is a model of stochastic states of nature from minus infinity to plus infinity in continuous time. As a model of egg counts, it is obviously wrong. Egg counts are integers bounded below by zero and above by the biological potential of Wandering Albatross. Eggs are laid once a year.

On reflection, every state of nature has lower and upper bounds. We mostly ignore them. When time is not important, we are willing to use the Normal Distribution as an approximation, even though there are many alternative distributions with lower and upper bounds. When states of nature are discrete, we are willing to use the Normal Distribution as an approximation of the Binomial Distribution. The Ornstein-Uhlenbeck Process generalises the Normal Distribution to include time.

We have not yet found solutions for stochastic processes with lower and upper bounds. We might use Geometric Brownian Motion, instead. It has an absorbing boundary at zero, has no upper bound and actively diverges toward infinity. According to Geometric Brownian Motion, Wandering Albatross would go extinct if no eggs were laid and could, and probably would, lay an infinite number of eggs. An improvement might be the Gompertz Growth Process. It converges rather than actively diverging toward infinity. A further improvement might be the Square Root Process. It

can have a reflecting, rather than absorbing, boundary at zero and it converges.

Until we have an implementation of the Square Root Process, the Ornstein-Uhlenbeck Process is an approximation. It does not require that Wandering Albatross become extinct if they fail to lay eggs this year. True, it has no lower or upper bounds on the number of eggs. But it converges toward a long-term mean with a long-term variance.

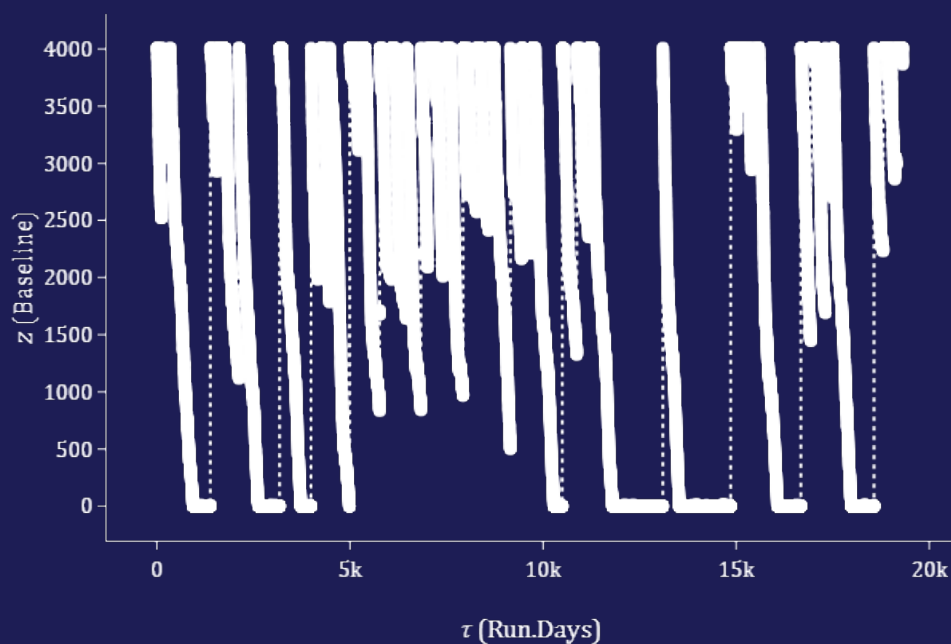
See Also: Recovery

Farm Dams and Destocking

Australia is thirsty country. Being a sheep is thirsty work. If the dam runs dry, a sheep is in trouble. Farmers know to anticipate trouble and destock the paddocks.

The destocking decision is a tipping point reached sometime before the dam is dust. Research by Clayton *et al.* evaluated windbreaks to suppress evaporation from farm dams in the Riverina and delay reaching the tipping point.²

An integrated hydrological and merino flock model was developed and simulated in daily time steps. The plot below shows water volumes in the dam for the Baseline without windbreaks. The vertical axis has the cubic metres of water and the horizontal axis has days in the simulation run.



The water in the dam is obviously bounded below at zero and above at 4,000 cubic metres. The Ornstein-Uhlenbeck Process has no lower or upper bounds. Is it an appropriate model of water volumes?

Geometric Brownian Motion and the Gompertz Growth Process have an absorbing lower bound and no upper bound. According to these processes, if the dam ever goes dry, it will never fill again. The Wright-Fisher model of population genetics has

²Helena Clayton, Sally Thompson, Tim Capon, Greg Hertzler, Philip Graham, and David Linden mayer, 2026,'Stock water management to improve drought resilience in intensive-use grazing land scapes', corresponding author: tim.capon@csiro.au

absorbing lower and upper bounds. We don't know how to solve this model, but even if we did, it wouldn't be appropriate. If the dam ever goes dry it would never fill, and if it ever fills it would never go dry. A Square Root Process can have a reflecting boundary at zero, allowing the dam to fill again, but it has no upper bound. Until we discover and solve a stochastic process with reflecting lower and upper bounds, the Ornstein-Uhlenbeck Process is an approximation of water volumes.

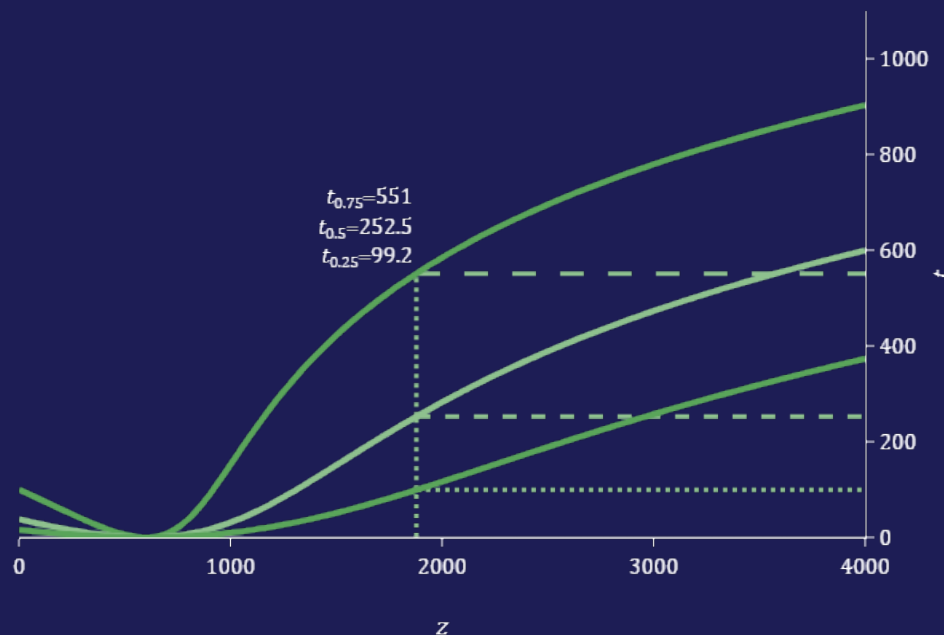
The table below contains the estimated Ornstein-Uhlenbeck parameters for three scenarios: the Baseline, a 20% reduction in evaporation and a 40% reduction in evaporation.

Ornstein-Uhlenbeck parameters				
	rho	mu	sigma	alpha
Baseline	0.00295478	1878.55	117.009	0.997054
20% scenario	0.00255145	2038.78	107.962	0.997455
40% scenario	0.00223287	2246.19	98.9761	0.997772

The alpha parameter is almost 1 for all scenarios. Water volumes are almost a random walk. Not quite, however, because the scenarios converge slowly at rates rho toward long-term means mu, which increase with reduced evaporation. The long-term variances are calculated as $\sigma^2/(2\rho)$. Because rho is small, the long-term variances are large, but they decrease with reduced evaporation.

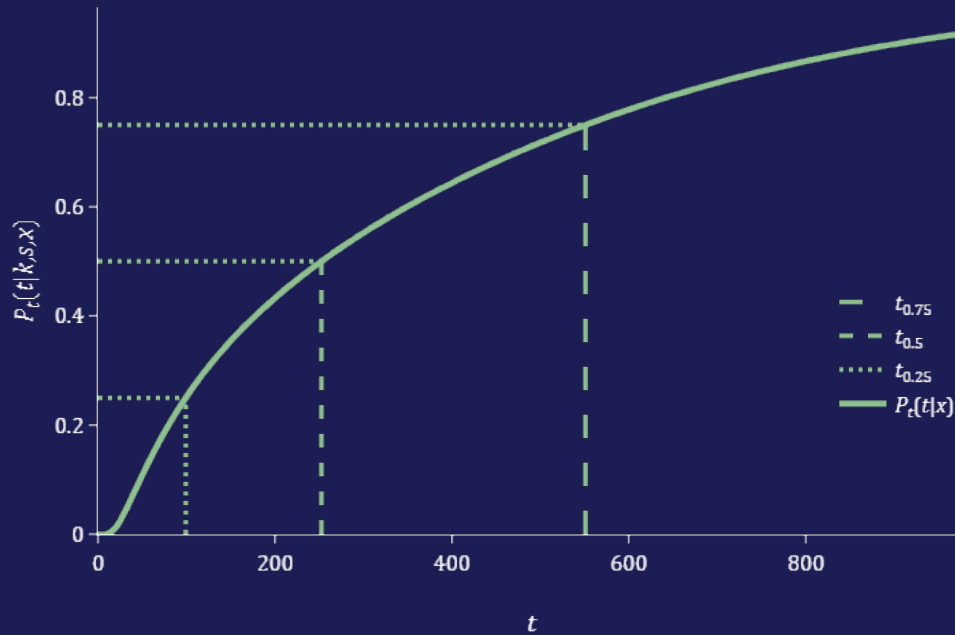
Based on farmers' experience, the tipping point is three months of water supply, which translates to 600 cubic metres. When the tipping point is reached, farmers begin destocking. Destocking changes the sheep flock and is expensive to reverse. Times to an irreversible tipping point are measured as First Passage Times.

The plot below shows First Passage Times for the Baseline to reach a water volume of 600 cubic metres. All possible water volumes are on the horizontal axis. Times are on the vertical axis as percentiles: 25%, 50% and 75%, where 50% is the median.



If the initial water volume is the long-term mean of 1,879 cubic metres, there is a 25% chance that water volumes will reach the tipping point before 99 days, a 50% chance before 252 days and a 75% chance before 551 days.

The plot below shows the First Passage Time Probabilities on the vertical axis for all possible First Passage Times on the horizontal axis. This plot is for an initial water volume of 1,879 cubic metres. Different initial water volumes have different probabilities.



The table below has median days to reach the tipping point for the three scenarios and for selected initial water volumes.

Median days to destocking			
Initial volume (m ³)	Baseline	20% scenario	40% scenario
1,879	252	316	406
2,039		363	464
2,246			531

In the first column, initial water volumes are the long-term means of the three scenarios. The other columns are the median days to reach the tipping point.

Starting from an initial volume of 1,879 cubic metres, the Baseline will have 252 days. The 20% scenario will have 316-252 = 64 more days and the 40% scenario will have 406-252 = 154 more days. For the same initial water volume, reduced evaporation increases the days to reach the tipping point.

Reduced evaporation will also increase the water volume in the dam. If the Baseline starts from 1,879 cubic metres and the 20% scenario starts from 2,039 cubic metres, the 20% scenario will have 363-252 = 111 more days. If the 40% scenario starts from 2,246 cubic metres, it will have 531-252 = 279 more days—more than double the time to reach the tipping point.

Windbreaks can delay destocking and increase water volumes. The tipping point of 600 cubic metres is chosen from farmers' experience without windbreaks. Further

research will understand the benefits and costs of destocking and the cost of planting windbreaks. New tipping points will be calculated as Decision Thresholds in a Real Options analysis.

See Also: Tipping Point

Agriculture and Climate

Are we approaching our Easter Island moment? Is our ability to exploit our environment outpacing our ability to conserve it? Will our legacy be a few stoic statues while the Earth spins on?

In 1798, Malthus predicted disaster. In 1972, 'Limits to Growth' predicted we had 30 years left. In 1992, 'Beyond the Limits', in 2004, 'The 30-Year Update', in 2012, 'A Global Forecast', and in 2022, 'Limits and Beyond' warned us disaster looming.

Economists are skeptical—citing the evidence that the Earth may be spinning but we are still clinging on. Obviously, our technology to exploit has not out paced our ability to adapt. Malthusian scarcity has been replaced with Ricardian scarcity. Limits imposed by nature have been substituted with our ability to bid scarce resources into higher value uses. There are no limits to growth.

Maybe, but the pace is picking up. How fast can we adapt? Malthus predicted the first limit will be land suitable for agriculture. 'Limits to Growth' predicted the same just before the Green Revolution proved them wrong—so far. Will agriculture adapt fast enough to a changing climate or will we go hungry?

Hertzler *et al.* used Real Options to evaluate how Australian broadacre agriculture will adapt to climate change.³

As with many studies, spatial analogues were a proxy for a changing climate. Favourable locations were assumed to become like less favourable locations. For South Australia, a climate transect was chosen to straddle Goyder's Line, a line drawn in the 1860's to demarcate land suitable for grazing from land suitable for cropping. For New South Wales, three locations were chosen, varying from wheat dominant through mixed cropping to sheep dominant agriculture. For Western Australia, rain-fall iso hyets were used.

For South Australia and New South Wales, crop yields under climate variability were simulated by the Agricultural Production Systems Simulator (APSIM). Sheep stocking rates were simulated using the GrassGro model. Wheat yields and sheep stocking rates were converted to gross margins using data on representative costs and returns. For Western Australia, actual farm data was used.

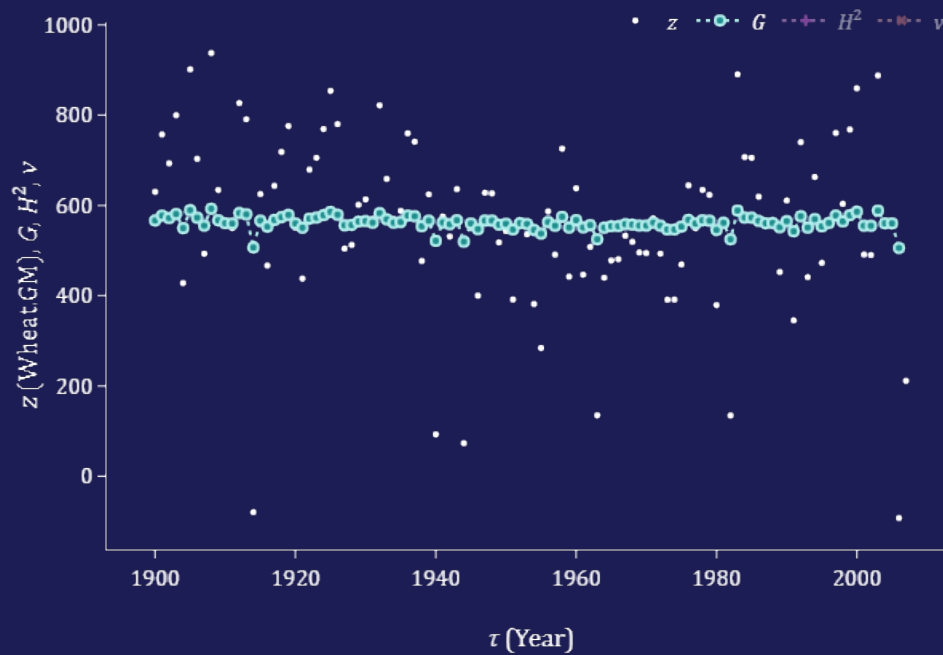
Sanderson *et al.* presented the methods and summarised results for South Australia.⁴

Here, we revisit South Australia and New South Wales and improve the method for estimating Ornstein-Uhlenbeck Processes. We only revisit the main results. There

³Hertzler, G, Sanderson, T, Capon, T, Hayman, P, Kingwell, R, McClintock, A, Crean, J, Randall, A 2013 *Will primary producers continue to adjust practices and technologies, change production systems or transform their industry – an application of real options*, National Climate Change Adaptation Research Facility, Gold Coast, pp. 93, <https://nccarf.edu.au/will-primary-producers-continue-adjust-practices-and-technologies-change-production/>

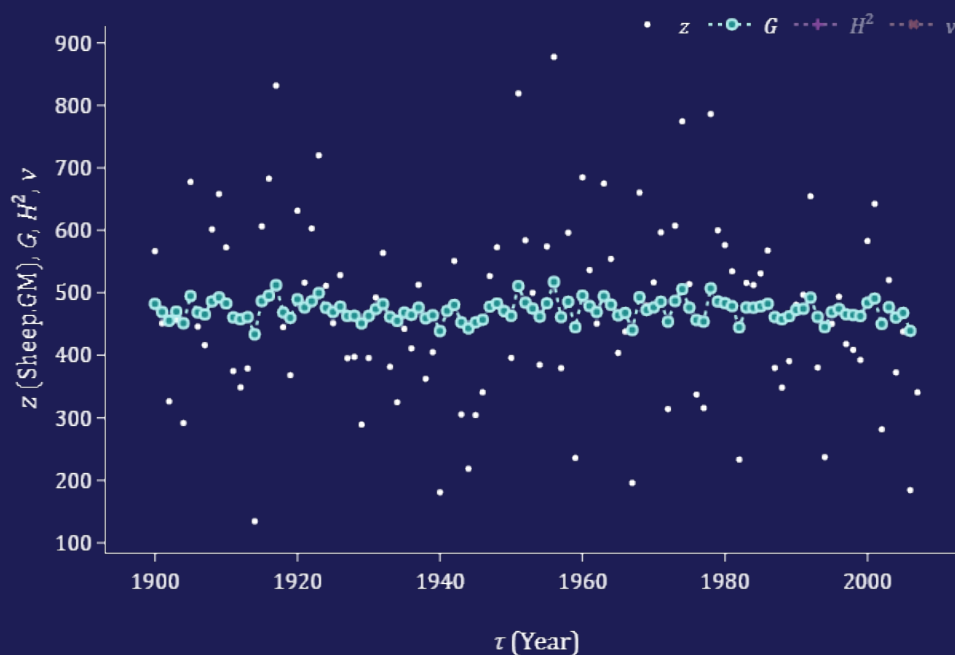
⁴Todd Sanderson, Greg Hertzler, Tim Capon, and Peter Hayman, 2015, 'A real options analysis of Australian wheat production under climate change,' *Australian Journal of Agricultural and Resource Economics*, 60, pp. 79-96. <https://doi.org/10.1111/1467-8489.12104>

is much more in the studies. The plot below shows estimated wheat gross margins at Clare, which sits on the good side of Goyder's Line.



The horizontal axis is years. The vertical axis is wheat gross margins which are almost stationary over time. This year's gross margin is little help in predicting next year's gross margin.

The plot below shows estimated sheep gross margins.



Sheep gross margins, also, are almost stationary.

Regimes of Wheat and Sheep are simulated for locations Clare, Orroroo and Hawker along a climate transect bisecting Goyder's Line. The table below shows parameter estimates for gross margins.

Ornstein-Uhlenbeck parameters for South Australia						
Regime	Location	rho	mu	sigma	alpha	CV
Wheat	Clare	2.47353	560.904	418.673	0.503552	0.34
	Orroroo	19.23600	432.486	2017.900	0.500000	0.75
	Hawker	2.81141	293.031	686.514	0.501807	0.99
Sheep	Clare	2.18175	471.584	308.076	0.506367	0.31
	Orroroo	2.56019	186.622	210.178	0.502987	0.50
	Hawker	2.21838	157.649	234.85	0.505917	0.71

Parameter alpha close to $\frac{1}{2}$ confirms that gross margins are almost stationary for both regimes at all locations. According to long-term means, mu, Wheat is expected to be more profitable than Sheep. Expected gross margins decrease along the climate transect from Clare to Orroroo to Hawker.

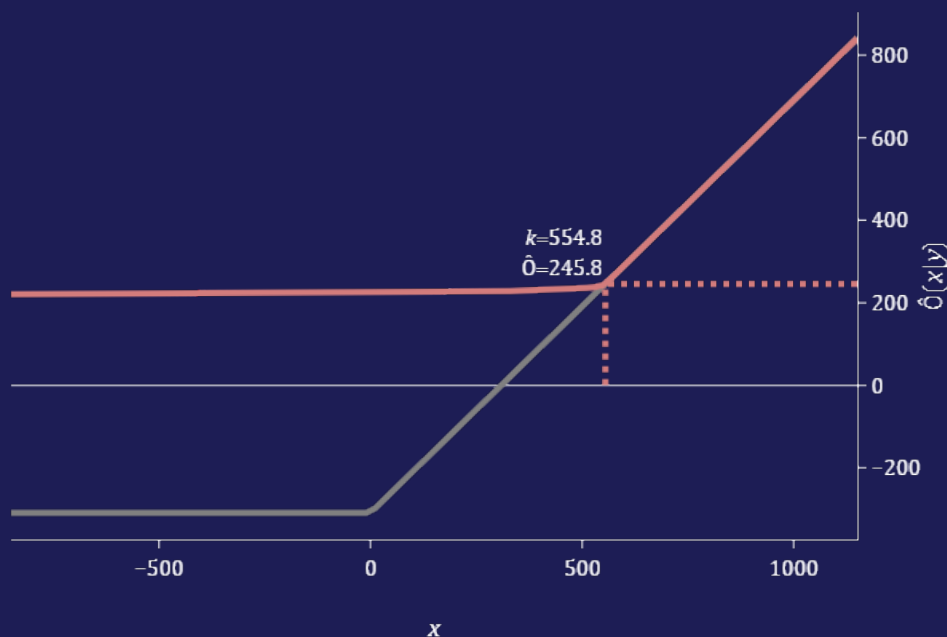
Variability is harder to judge. The long-term variance is $\sigma^2/(2\rho)$. Dividing the square root of the long-term variance by the long-term mean gives the Coefficient of Variation (CV). Judging by the CV's, Wheat is more variable than Sheep. Variability is manageable at Clare, extreme at Orroroo and unbelievable at Hawker.

The variable costs are subtracted in the gross margins, but the start up costs to enter and the salvage values upon exit are in the table below.

Start Up Costs and Salvage Values

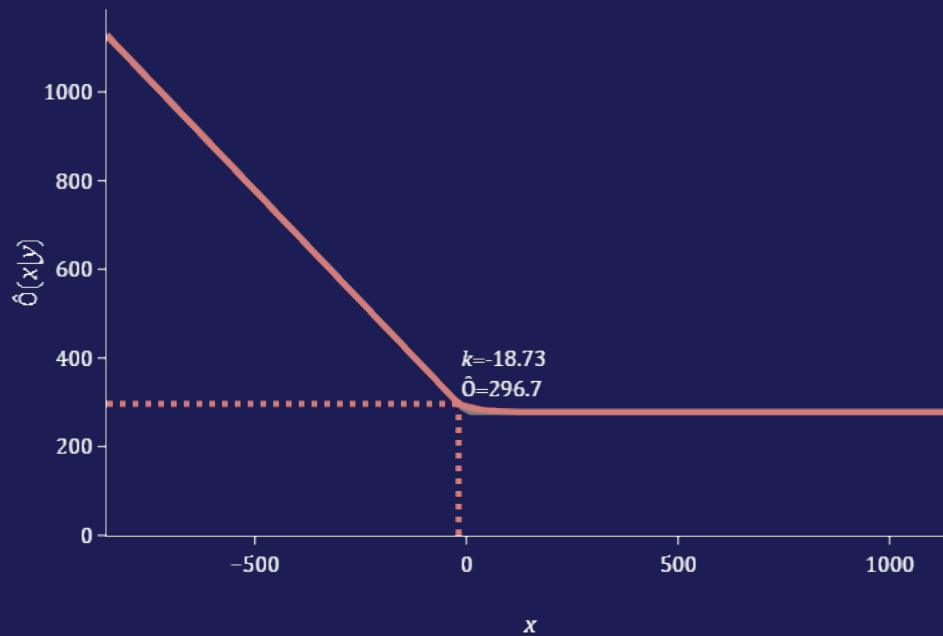
Regime	Entry	Exit
Wheat	309	278
Sheep	32	29

As an example, the plot below shows the Decision Threshold to enter the Wheat regime at Clare.



The Payoff function and Option Prices are shifted down by the start up costs of \$309. Hence, Decision Threshold k minus Option Price $\hat{O}$ is \$309.

The plot below shows the Decision Threshold to exit the Wheat regime.



The Payoff function and Option Prices are shifted up by the salvage value of \$278. Hence, Decision Threshold k plus Option Price $\hat{O}$ is \$278.

The table below has the entry and exit Decision Thresholds, k , and the corresponding Option Prices, $\hat{O}$.

Decision Thresholds for South Australia					
Regime	Location	Entry		Exit	
		k	$\hat{O}$	k	$\hat{O}$
Wheat	Clare	554.8	245.8	-18.7	296.7
	Orroroo	447.8	138.8	-63.7	341.7
	Hawker	320.4	11.4	-66.4	344.3
Sheep	Clare	459.7	427.7	-14.0	43.0
	Orroroo	183.3	151.3	-13.2	42.2
	Hawker	157.9	125.9	-20.7	49.7

For Entry, k minus $\hat{O}$ equals the start up costs. For Exit, k plus $\hat{O}$ equals the salvage value. Start up costs decrease the value of an Entry Option. Salvage value increases the value of an Exit Option. Entry Options for Sheep are more valuable than Entry Options for Wheat due to low start up costs. Exit Options for Sheep are less valuable due to low salvage values.

At Clare, k is below the long-term mean, μ . At Orroroo and Hawker, k is at or above μ . Greater uncertainty delays entry into both Wheat and Sheep. The value of Entry Option $\hat{O}$ decreases from Clare to Orroroo to Hawker. Conversely the value of Exit Option $\hat{O}$ increases.

If climate change transforms Clare into Orroroo, the option to grow Wheat will be worth \$107.0 per hectare per year less and the option to grow Sheep will be worth

\$276.4 per hectare per year less. If Orroroo becomes like Hawker, the option to grow Wheat will be worth \$127.38 per hectare per year less and the option to grow Sheep will be worth \$25.40 per hectare per year less.

Hertzler *et al.* analysed four regimes for New South Wales. The table below has results for only Wheat and Sheep.

Decision Thresholds for New South Wales					
Regime	Location	Entry		Exit	
		k	$\hat{O}$	k	$\hat{O}$
Wheat	Cootamundra	447.7	138.7	-33.1	311.1
	Temora	319.1	10.1	-56.1	334.1
	Narrendera	309.0	0.0	-59.2	337.2
Sheep	Cootamundra	155.2	123.2	-15.7	44.7
	Temora	82.6	50.6	-13.1	42.1
	Narrendera	45.4	13.4	-20.1	49.1

Along the climate transect from Cootamundra to Temora to Narrendera, the value of Entry Options for Wheat go to zero. The value of Entry Options for Sheep become very small. As Cootamundra becomes like Temora and Temora becomes like Narrendera, farmers will exit from Wheat production, but they may not enter Sheep. They may cash out and go fishing.

In summary, an Entry Option values the flexibility to choose whether or not to enter. It includes both the Obligation and the Exit Option. The Obligation equals the actual benefits minus costs. The Exit Option values the flexibility to choose whether or not to exit.

The Entry Option at the Decision Threshold is the cash rent for farmland under uncertainty. The Entry Option capitalised is the purchase price. With climate change, the price of a hectare of farmland will be thousands of dollars less.

Spatial analogues for climate change are used in many hedonic pricing studies of farmland prices. The Sanderson *et al.* and Hertzler *et al.* studies are an alternative assessment which ascribe the changes in land values to changes in option values. Under uncertainty, farmers will be cautious about exercising their options to enter Wheat or Sheep. But even if they hold, rather than exercise, their options, the options are still valuable.

Hedonic pricing studies, and the results presented here, assume Ricardian scarcity. Wheat and Sheep are grown on land of different capabilities and are entered and exited independently. The Sanderson *et al.* and Hertzler *et al.* studies first consider Ricardian scarcity and then go on to consider Malthusian scarcity. Wheat and Sheep compete to occupy the same hectare of land and are entered and exited in a sequence of Real Options.

At this writing, sequences of Real Options can be analysed in the R6 console app and the Excel workbooks, but not in the RShiny app.

See Also: Entry Option, Obligation / Prohibition, Exit Option

Fisheries and ITQs

In the game of scams, the role of economists is to outwit the scammers—to devise policies that let people enjoy the fruits of their toil and the benefits from common property.

Scammers are crafty.

In stock markets they craft 'narrative stocks'. I bought a narrative called Blue Sky Alternative Investments. Whilst I wasn't looking, the narrators stole my money. The price per earnings ratio of my Tesla stock is over 300. I plan to live another 300 years and give Tesla the chance to repay my trust. If my Bitcoin becomes legal tender, I will invest in salt for the new barter economy. Hyperinflation will steal all my hard earned cash.

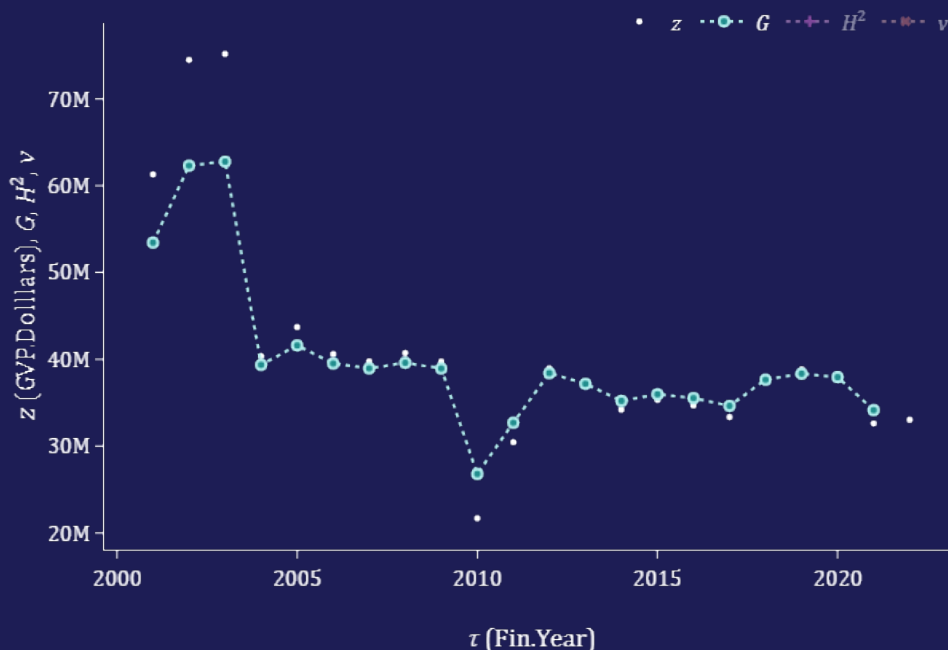
I need help, but economists are not much. Occasionally, they outwit bureaucrats.

When scammers crash the system, bureaucrats may panic. Economists can sneak in with their truth serums and conduct interviews. 'So you want to extract the peoples' property for your private benefit. Tell us truthfully about your costs of doing business.'

One recipe for truth serum is auctioning the rights to extract minerals and energy. This recipe made Norway rich. Another recipe is a cap and trade scheme. This recipe helped solve Los Angeles' smog problem. After its most valuable fisheries crashed, Australia began using this recipe. The cap and trade scheme for fisheries is Individual Transferable Quotas, or ITQs. ITQs work best for single species fisheries with high value fish such as Tropical Rock Lobsters and Southern Bluefin Tuna.

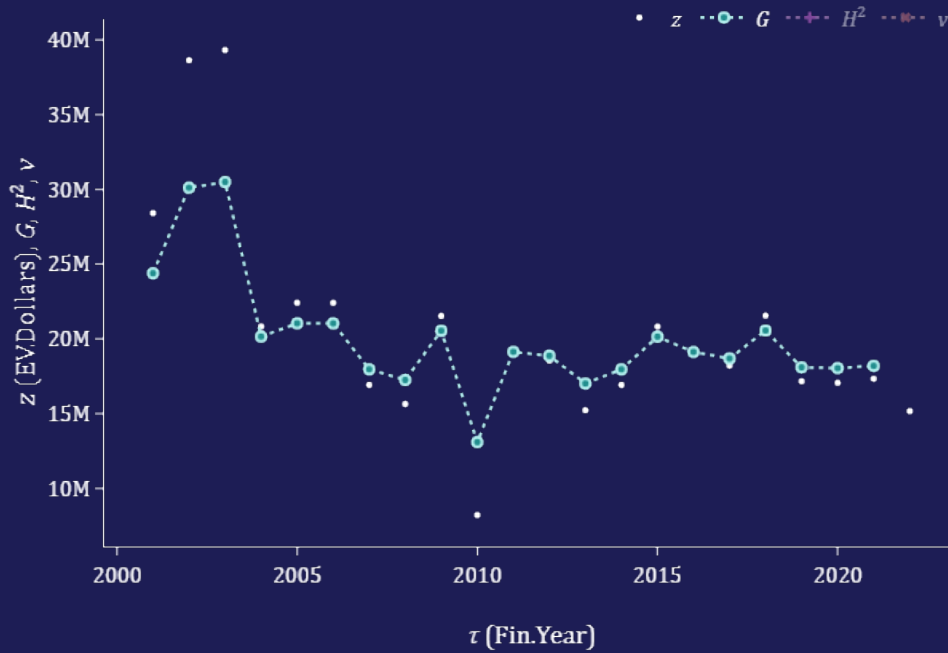
This case study investigates ITQs for Southern Bluefin Tuna under uncertainty. We use data from the National data for Australia's ecosystem services.⁵

There is extensive data for all fish species by region and habitat. For this investigation, we aggregate Southern Bluefin Tuna over all of Australia. The plot below shows the Gross Value Product of the Southern Bluefin Tuna catch on the vertical axis and the financial year on the horizontal axis.



Catch is multiplied by the market price to get the Gross Value Product. The plot below is for ITQs.

⁵Pascoe, Sean; Liu, Ning; Scheufele, Gabriela; Tetreault Campbell, Sally; & Schmidt, Becky (2024): National data for Australia's ecosystem services: fisheries biomass provisioning services (2000-01 to 2020-21). v9. CSIRO. Data Collection. <https://doi.org/10.25919/nzrp-6702>



The vertical axis is labeled EV for Ecosystem Values. These are measured in various ways for different species, but for Southern Bluefin Tuna these are the annual rental value of the ITQs.

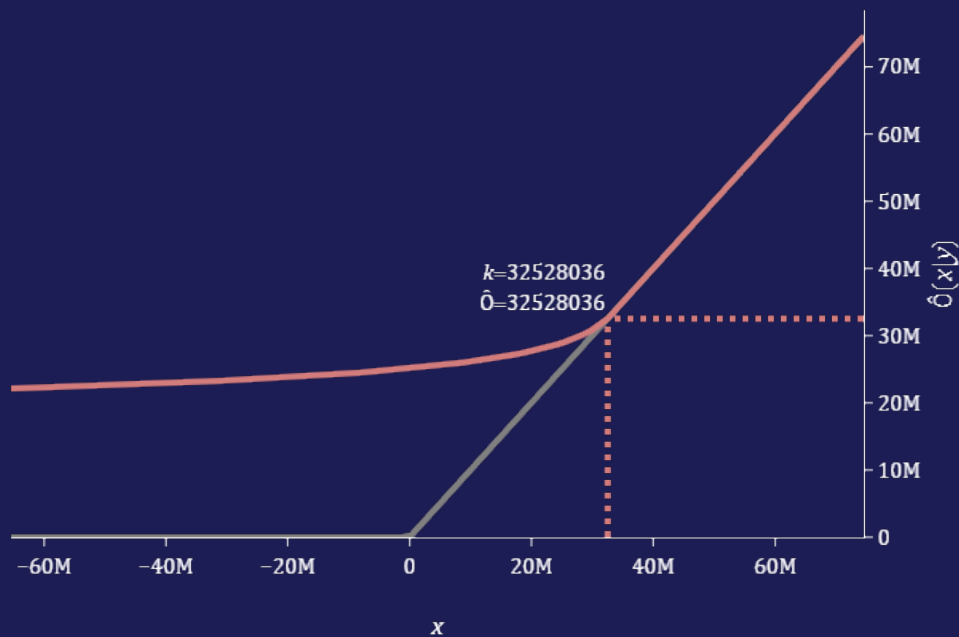
The table below contains the estimated Ornstein-Uhlenbeck parameters for the Gross Value Product and the Ecosystem Values.

Ornstein-Uhlenbeck parameters				
	rho	mu	sigma	alpha
Gross Value Product	0.396596	37280885	10313308	0.726199
Ecosystem Values	0.580743	19288105	7344241	0.65651

The alpha parameters identify both series as somewhere between stationary and a random walk. The rho parameters show that Ecosystem Values converge a bit faster and the sigma parameters show that Ecosystem Values are less uncertain. The mu parameters are the long-term means. In the long term, the Southern Bluefin Tuna fishery generates \$37.3 million per year. More than half of this value, \$19.3 million per year, is the ecosystem provisioning services it provides.

Fishers will complain about the imposition of ITQs. To forestall the complaints, fishers could be gifted the ITQs. New Zealand does it this way. Fishers would still have to comply with the quotas. Under uncertainty, would fishers choose to enter a fishery that restricts their catch?

The plot below shows the Decision Threshold if fishers are gifted the ITQs. Both the horizontal and vertical axes are Gross Value Product.

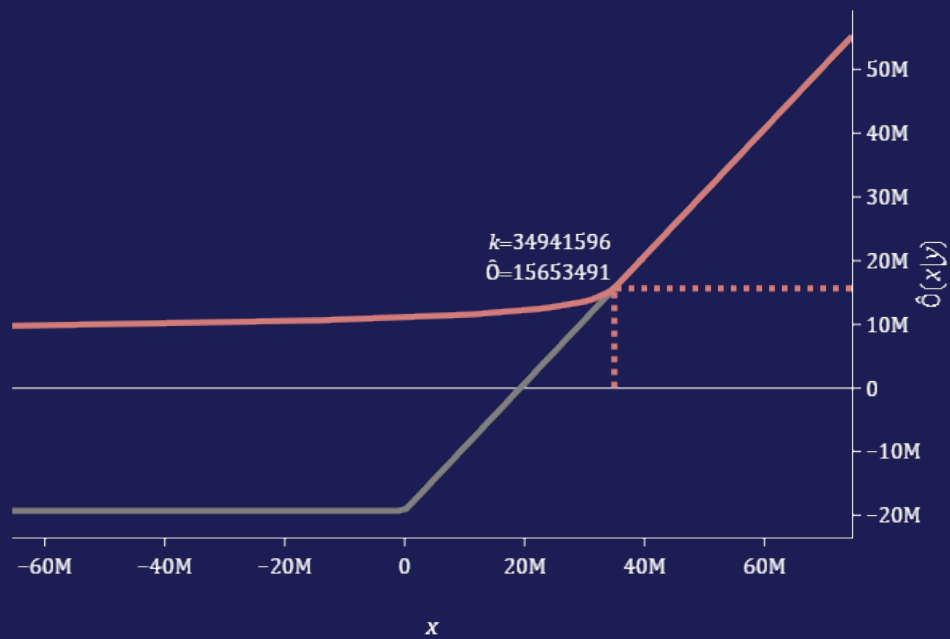


On the horizontal axis, the break even point is set to zero. Below zero, the fishery loses money. Above zero, the fishery makes money along the 45° line. The break even point will be increased later. The curve is the value of the Entry Option. It rises to equal the 45° line at the Decision Threshold.

The Decision Threshold, k , is \$32.5 million per year. Because the break even is set to zero, the Option Price at the Decision Threshold, $\hat{O}$, is also \$32.5 million per year. Fishers will enter the fishery before the Gross Value Product reaches its long-term mean of \$37.3 million per year.

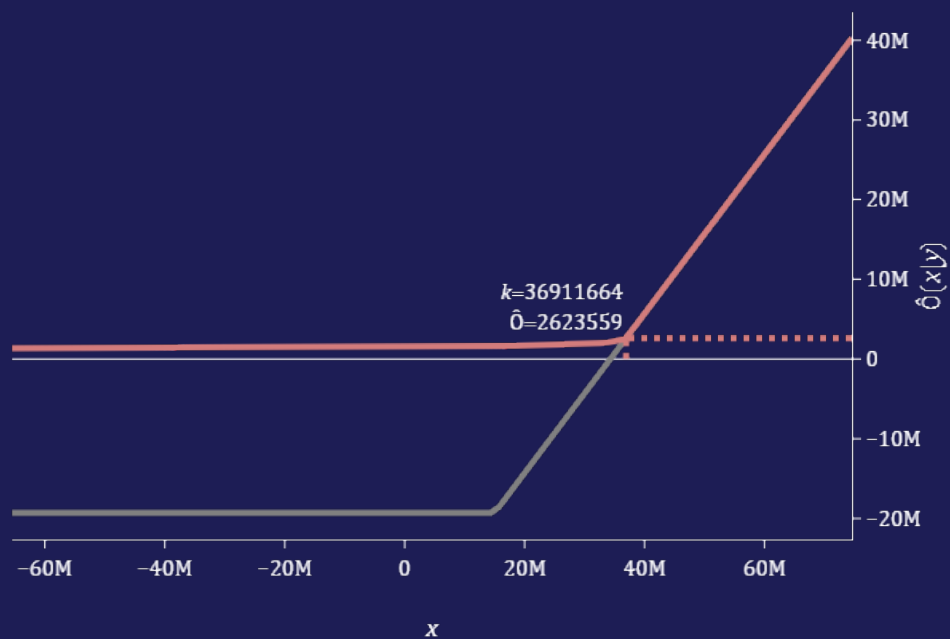
Suppose fishers are not gifted the ITQs. Suppose economists hold an auction, instead. We can imagine the outcry. Would fishers pay the rental on ITQs and enter the fishery?

The plot below shows the Decision Threshold if fishers rent the ITQs. Curves and lines are shifted down by the rental price of \$19.3 million per year.



The Decision Threshold is delayed slightly to $k = \$34.9$ million per year. The Option Price at the Decision Threshold is lower at $\hat{O} = \$15.7$ million per year. This is the value to fishers of the option to enter the fishery, without considering the costs of catching fish.

The plot below increases the break even point to \$15 million per year. Lines and curves are shifted right.



Fishers rent the ITQs and pay the costs of catching fish and still enter the fishery.

Over the long term, fishers of Southern Bluefin Tuna pay 52% of their Gross Value Product to rent ITQs. A similar analysis shows fishers of Tropical Rock Lobsters pay 43% of their Gross Value Product to rent ITQs.

Policies for most of Australia's resources are *ad valorem* royalties. Royalties for Queensland coking coal were recently raised to around 23% of revenue. Royalties for Western Australia iron ore are 7.5% of revenue. Royalties for gold are 2.5%. Every decade or so, the federal government proposes resource rent taxes, presumably because the federal government collects taxes but not royalties. Every so often, industry pundits ask, 'Whatever would state governments do without royalties?'

A more interesting question is, 'What could state governments do with the money if economists injected some truth into the discourse by holding auctions?'

See Also: Entry Option, Resource Rent

Carbon Contracts

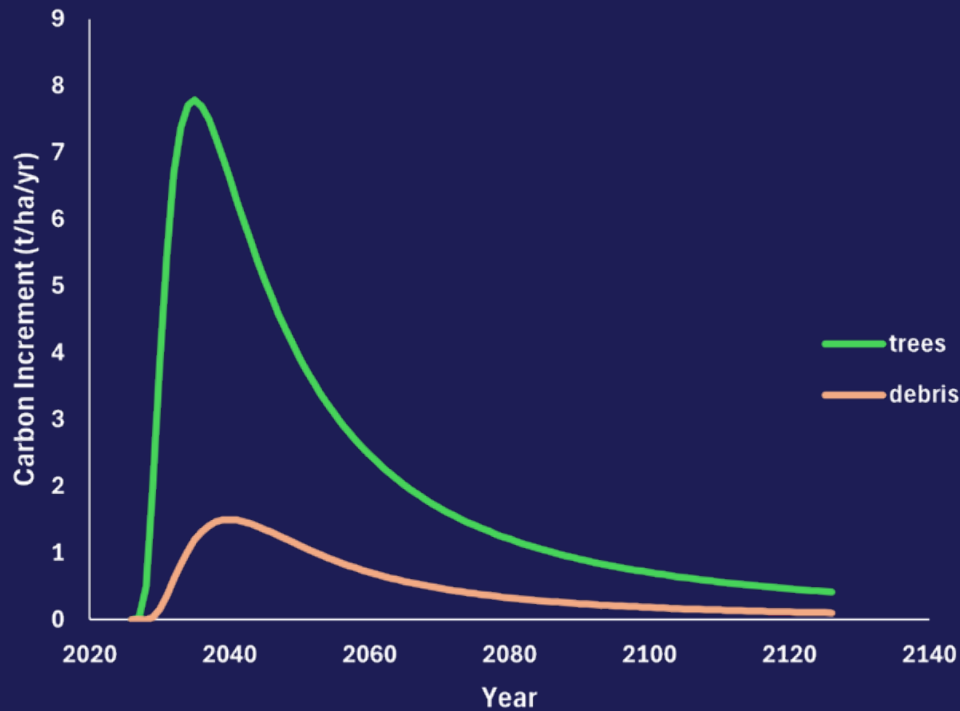
Diamonds are forever—sequestered carbon not so long. Australian Carbon Credit Units (ACCU's) are tonnes of sequestered carbon with a 'permanence period' of either 100 years or 25 years. ACCU's are a commodity to produce and sell.⁶

My home is in the Jarrah forest. We watch the trees die from the stress of humans living amongst them. The shire has strict rules about firebreaks, clearing around houses and fuel reduction. We, and our neighbours, routinely clear and burn. Our neighbours sink bores to extract groundwater for their gardens and vineyards. Mining companies explore for bauxite. Bauxite is the sponge that holds the water.

If I produce ACCU's, can I offset the incentives to clear the forest? Mature Jarrah trees live 4 or 5 centuries. One of our trees has lived at least that long. A 100 year permanence promise seems too short for Jarrah. A 25 year promise seems too long for humans. I must promise one or the other to sell ACCU's.

Before planting, ACCU's are calculated with FullCAM (Full Carbon Accounting Model). The plot below is the output of FullCAM showing the increments of carbon I can produce by planting one hectare of Jarrah forest.

⁶Australian Government, Clean Energy Regulator (6 May 2025). <https://cer.gov.au/schemes/australian-carbon-credit-unit-scheme>

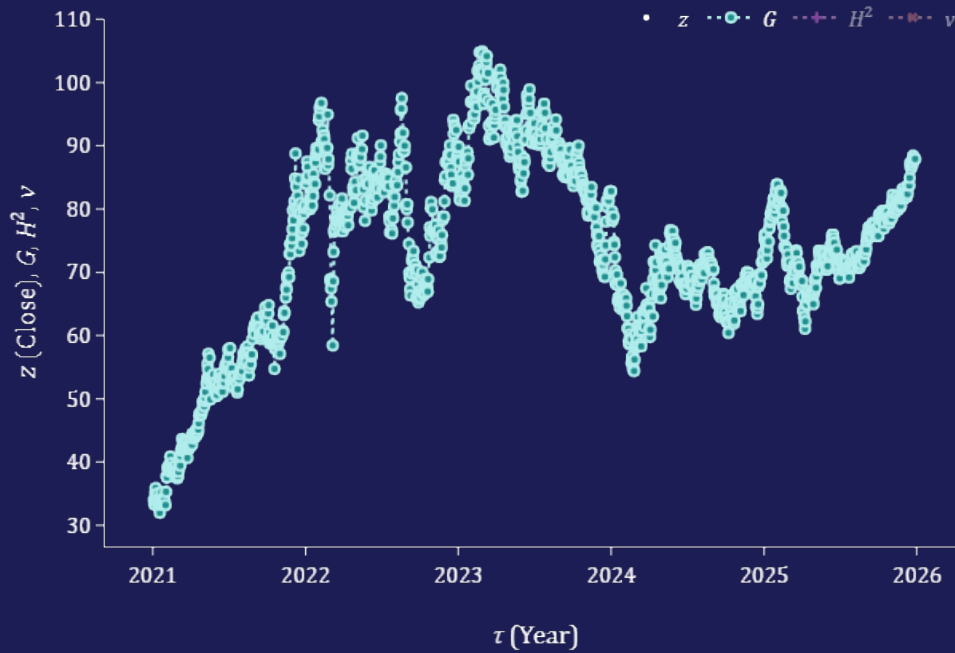


The vertical axis is increments of carbon in tonnes per hectare per year. The horizontal axis is years. By law, we are required to remove or burn the debris, leaving just the carbon in the trees. Obviously, I should enter a short-term contract, plant the trees, cut them down in a few years and plant again. But if I do that, I forfeit all my ACCU's. If I choose a 25 year permanence promise, I will be paid for only 80% of the carbon in the trees.

My search for prices of ACCU's and costs of planting and maintaining trees found snippets of information. But, as they say, deficient data does not deter a determined analyst.⁷

The European Union Emission Trading System is the longest running carbon trading scheme and collects data on prices. The plot below shows the Maximum Likelihood Estimation for closing market prices of a tonne of carbon.

⁷European Commission, Energy, Climate Change, Environment, EU Emissions Trading System. https://climate.ec.europa.eu/eu-action/carbon-markets/eu-emissions-trading-system-eu-ets_en
Data. <https://seekingalpha.com/symbol/EEEXM>



On the plot, prices look to be Brownian Motion or, in other words, a Random Walk. The table of estimates below confirms this.

	rho	mu	sigma	alpha
Parameters	2.84933	77.1144	30.8359	0.990746

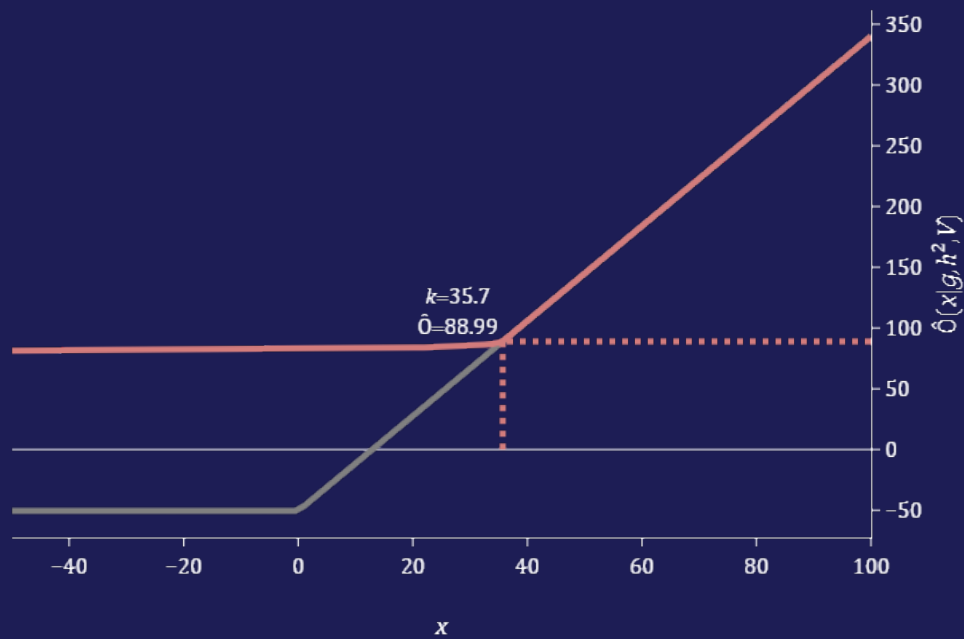
At first glance, the rate of convergence, rho, looks large. But prices are reported daily and time is in years. Each observation is over $1/365$ years and $\rho / 365$ is a small number. An alpha parameter close to 1 confirms that prices are almost, if not quite, Brownian Motion. Prices converge slowly to long-term mean mu of €77.11. If the Jarrah forest was in Europe, carbon sequestration would be more profitable. Unfortunately, the only publicly available price for an ACCU is a spot price of \$36.25 per tonne. Not to be deterred, I substitute 36.25 for mu.

I have a shovel and know how to dig holes, but will need to purchase Jarrah seedlings. I intend to plant 1,000 stems per hectare. There are several local nurseries and I expect seedlings will only cost \$1.00 per stem. We are required by law to maintain fencing and tend the forest and will subsidise any other establishment costs.

Putting it all together is a bit tricky. The ACCU's are an increasing and decreasing schedule of tonnes per hectare per year, the price of ACCU's is a stochastic process in dollars per tonne and the cost of seedlings is a lump sum in dollars per hectare. I won't start planting until the price of ACCU's reaches a Decision Threshold, as yet to be determined.

The schedule of tonnes per hectare per year can be converted to annuities. At an interest rate of 5%, the first 25 years of the schedule are equivalent to an annuity of 4.67473 tonnes per hectare per year. The first 100 years are equivalent to an annuity of 3.90497 tonnes per hectare per year. The annuity over 25 years must be multiplied by 80% to get 3.73978 tonnes per hectare per year. The establishment costs, converted to an annuity, are \$70.9525 per hectare per year over 25 years and \$50.3831 per hectare per year over 100 years.

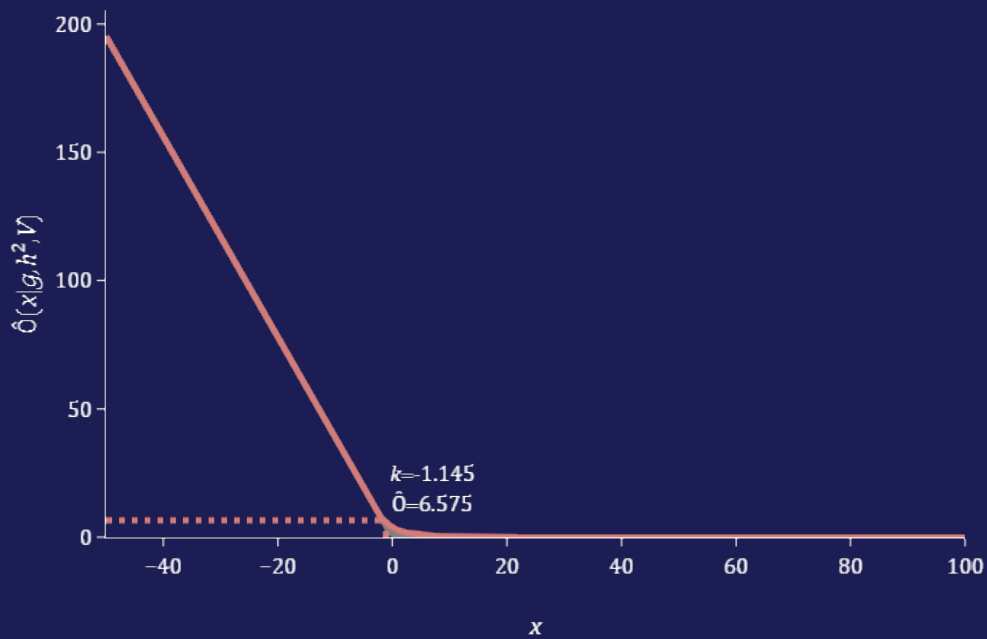
The plot below is the Decision Threshold for a 100 year permanence promise.



The horizontal axis is the price of an ACCU in dollars per tonne of carbon. The vertical axis is dollars per hectare per year. The payoff is shifted down by the establishment costs of \$50.381 per hectare per year. The slope of the payoff is the annuity over 100 years of 3.90497 tonnes per hectare per year. Multiplying by the price gives dollars per hectare per year.

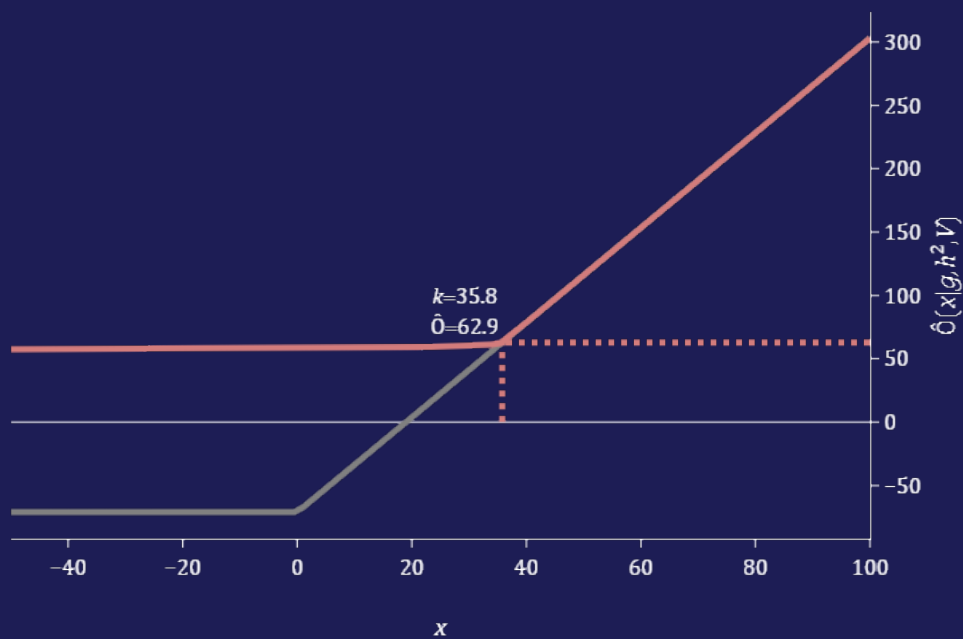
The Decision Threshold, k , is a price of \$35.7 per tonne, slightly less than the current spot price. The profit foregone while waiting for the Decision Threshold is Option Price $\hat{O}$ of \$88.99 per hectare. If I choose the 100 year option, I should enter now and expect to net around \$90 per hectare per year.

The 100 year promise eliminates the option to exit early. There is no Exit Option, at least in my lifetime. The plot below shows how much this might cost.



The payoff is not shifted up or down, assuming there are no costs or benefits other than those for producing ACCU's, an assumption to be questioned later. The slope of the payoff is -3.90497 tonnes per hectare. At the entry price of \$35.7 per tonne, the Exit Option is worth 3¢ per hectare. The value of the Exit Option is less than rounding error and will not affect the decision to adopt the 100 year promise.

I might make a 25 year promise, instead. The plot below shows the Decision Threshold.



The horizontal axis is the price of an ACCU in dollars per tonne and the vertical axis is dollars per hectare. The payoff is shifted down by the establishment costs of \$70.9525

per hectare and the slope is the annuity over 25 years of 3.73978 tonnes per hectare per year.

The Decision Threshold, k , is \$35.8 per tonne and the Option Price at the threshold is $\hat{O}$ of \$62.9 per hectare. Compared with the 100 year promise, the 25 year promise is adopted at a slightly higher price and the value of the option to enter is \$25.09 per hectare less.

I choose the 100 year promise and will order seedlings today.

What will happen in 100 years? According to FullCAM, my hectare of Jarrah forest will have 224 tonnes of carbon. Jarrah is Australia's premium furniture wood. It is excellent construction timber. Chopped as firewood, it currently sells for \$400 per tonne at my landscape supplier and \$1,500 per tonne at my local petrol station.

In today's money, after subtracting the costs of sharpening the chainsaw, my hectare of Jarrah forest will be worth \$80,000 as firewood. Putting the money in the bank at 5% interest will return \$4,000 per hectare per year.

At 100 years of age, my hectare of Jarrah forest will produce 0.42 ACCU's per year. In today's money, producing and selling ACCU's will return about \$14 per hectare per year.

Maybe I'll rethink. I am not an enthusiastic digger.

See Also: Contracts to Deliver

Yield Index Insurance

Where it exists, crop insurance is heavily subsidised. Attempts at commercially viable crop insurance have been defeated by adverse selection, moral hazard and basis risk. Attempts at writing insurance on a central weather station with payoffs on farm, saw farmers with lower rainfall sign up early and often. This was adverse selection. Attempts at writing multi-peril crop insurance saw farmers forget to fertilise and spray for pests. This was moral hazard. Attempts at writing area wide insurance with payoffs determined by an average yield for the area, saw farmers uninterested. This was basis risk.

Hedging of prices has basis risk—but not much. The price is more or less the same for everyone. The price a farmer gets is less than the market price, but the difference is transport costs. Futures and options on futures at a central market make sense for almost everyone.

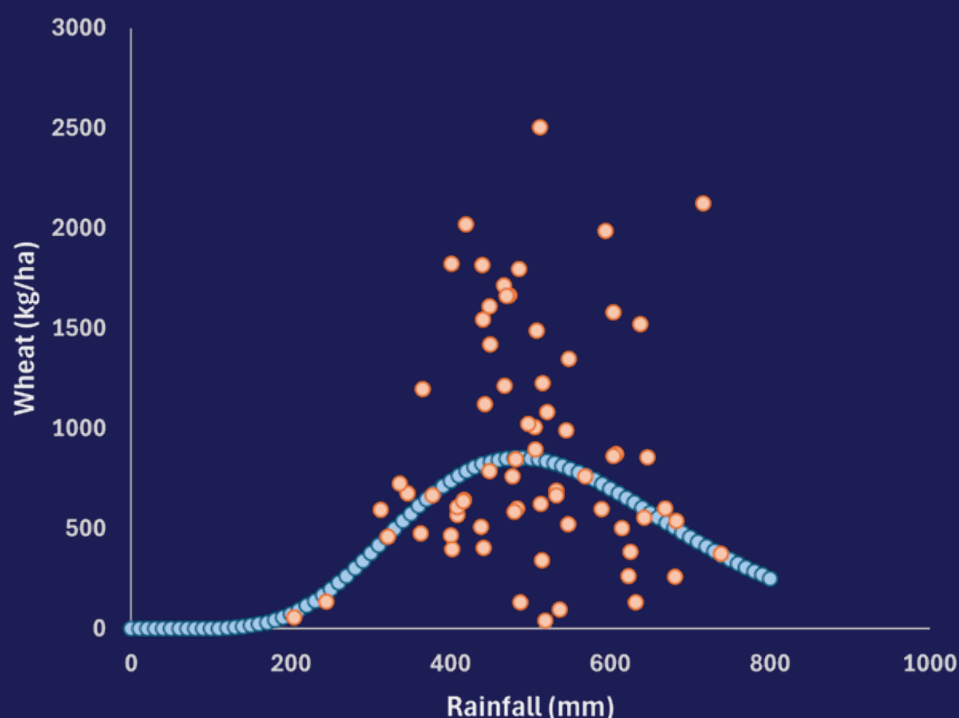
Hedging the weather has extreme basis risks. Weather at the airport may be uncorrelated with weather in a paddock. Crop yields also depend on soil types, nutrients, weeds and pests unrelated to distance from the weather station. Hedging the weather must be tailored to each paddock.

Weather stations are now cheap enough to put in every paddock, so long as the farmer doesn't put a lid on the rain gauge. A farmer could manage the nutrients and weeds and develop their own model of crop yields as a function of the weather. Each paddock could have its own yield function. An insurer could write contracts on these functions, with the payout determined by the recordings of the weather station.

In theory. Both farmers and insurers have better things to do. But this case study suggests how yields could be modelled and how yield index insurance could be priced. We will use a long time series of experimental data and weather records from the Waite Permanent Rotation Trial.⁸

⁸Sanderman, Jonathan; David, Rakesh; Moore, Andrew; Keith, Heather; & Farquharson, Ryan

There are several crop and pasture rotations in the data. We will use continuously cropped Wheat as a function of April through October growing season rainfall. The plot below has rainfall in millimetres on the horizontal axis and Wheat yields in kilograms per hectare on the vertical axis. An estimated yield index is plotted with the data.

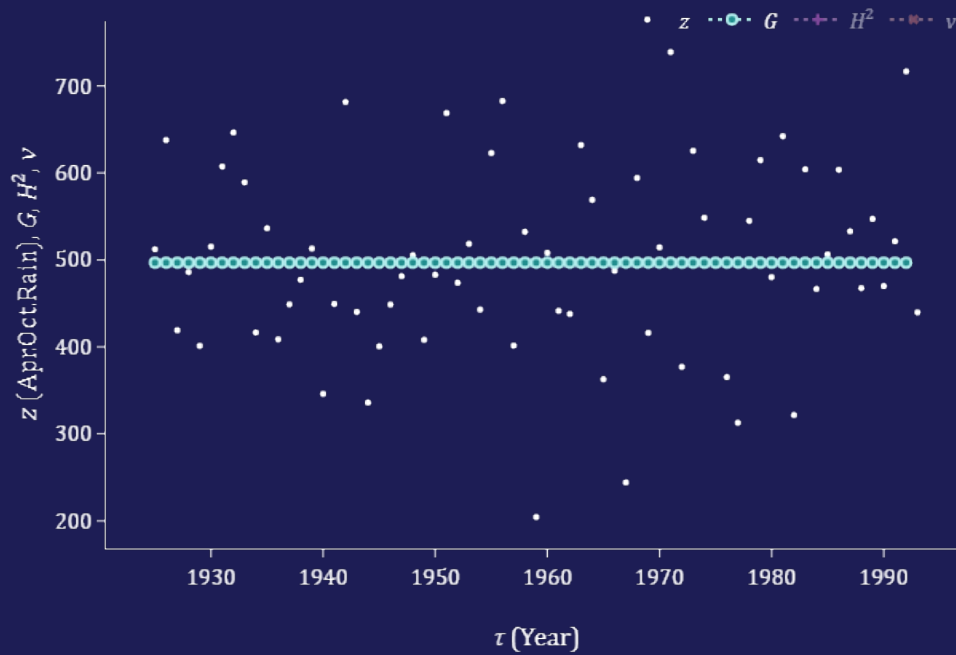


Obviously, many factors other than rainfall affected Wheat yields. These were not recorded, however. In some years, the rainfall was perfect but the yield was almost zero. A farmer would get no payout. Our Yield Index Insurance will have significant basis risks. To do better, we will need better data.

But let's continue. The estimated yield index has a maximum yield of 854 kilograms per hectare at a growing season rainfall of 485 millimetres. It has an inflection point at a rainfall of 316 millimetres. Later, we will use these numbers to specify the payout from insurance.

The plot below shows growing season rainfall as an Ornstein-Uhlenbeck Process.

(2015): Waite Permanent Rotation Trial. v4. CSIRO. Data Collection. <https://doi.org/10.4225/08/55E5165EC0D29>

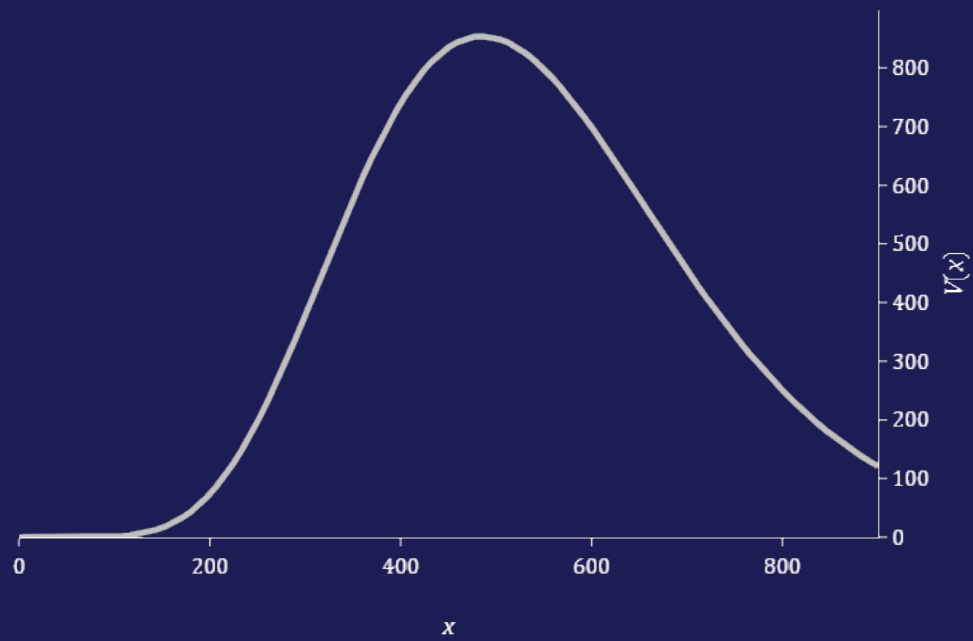


Rainfall appears to be stationary. The parameter estimates are in this table.

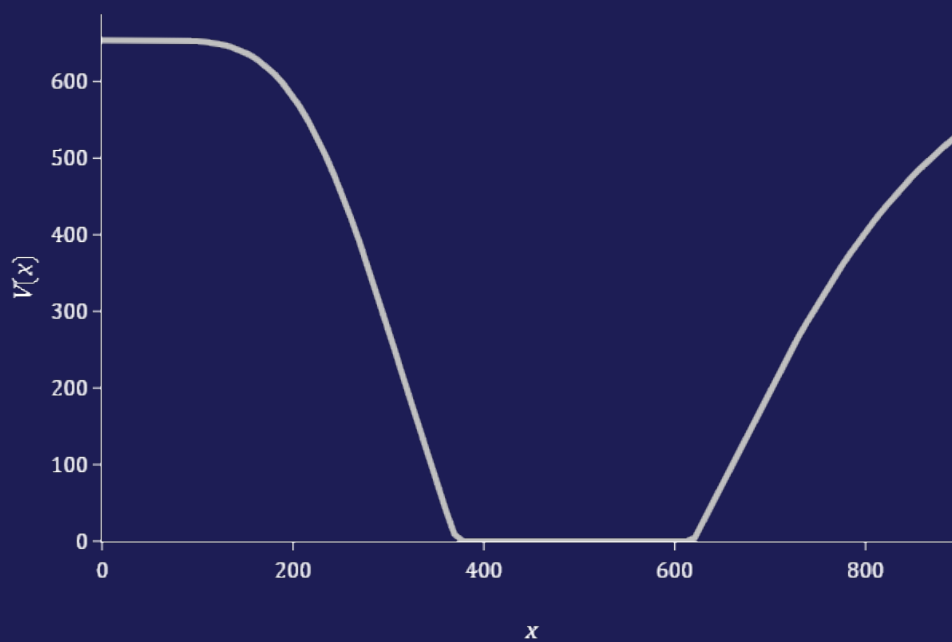
	rho	mu	sigma	alpha
Parameters	20.7286	496.756	701.125	0.5

A large rho and an alpha of $\frac{1}{2}$ confirm that expected growing season rainfall is its long-term mean of 497 millimetres. Last year's rainfall is of no help in predicting this year's rainfall.

Putting it all together, we will use the Finite Difference method to price our Yield Index. First we specify a terminal value with a maximum yield of 854 kilograms per hectare at a growing season rainfall of 485 millimetres and an inflection point at 316 millimetres of rainfall.

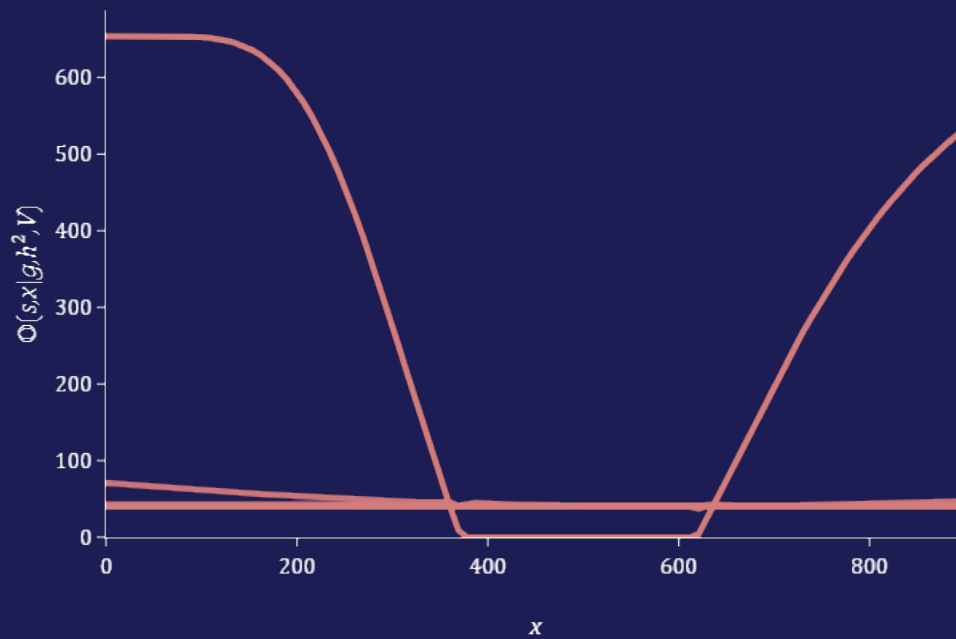


Then we chop off the top 200 kilograms of yield, flip it upside down and call it a Yield Index.



A Yield Index shows the damage. If growing season rainfall ends up below 378 millimetres or above 621 millimetres, the farmer receives payout. The payout is capped at 654 kilograms.

Because growing season rainfall is stationary, the price of Yield Index Insurance is almost constant.

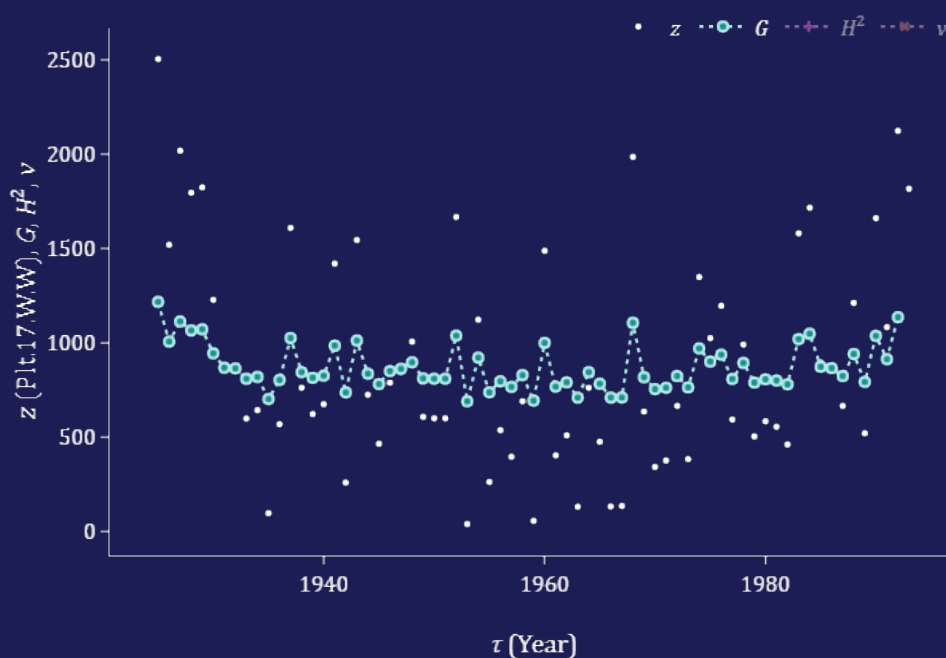


Purchased a year in advance, Yield Index Insurance will cost 39.98 kilograms per hectare, or about 4.7% of the maximum yield.

A farmer could decide what percentage of the crop they would be willing to pay for insurance and shift the yield index up or down. The payout need not be in dollars. The insurer could purchase wheat and deliver it for the farmer. More likely, the insurer and farmer will agree on a price per kilogram of wheat and settle in dollars.

Agriculture has entered the era of big data. Tillage and harvesting equipment collect data on the management and yield for each paddock. If a farmer agreed to share the data, an insurer might agree to sell them multi-peril crop insurance.

The plot below shows Wheat as an Ornstein-Uhlenbeck Process.

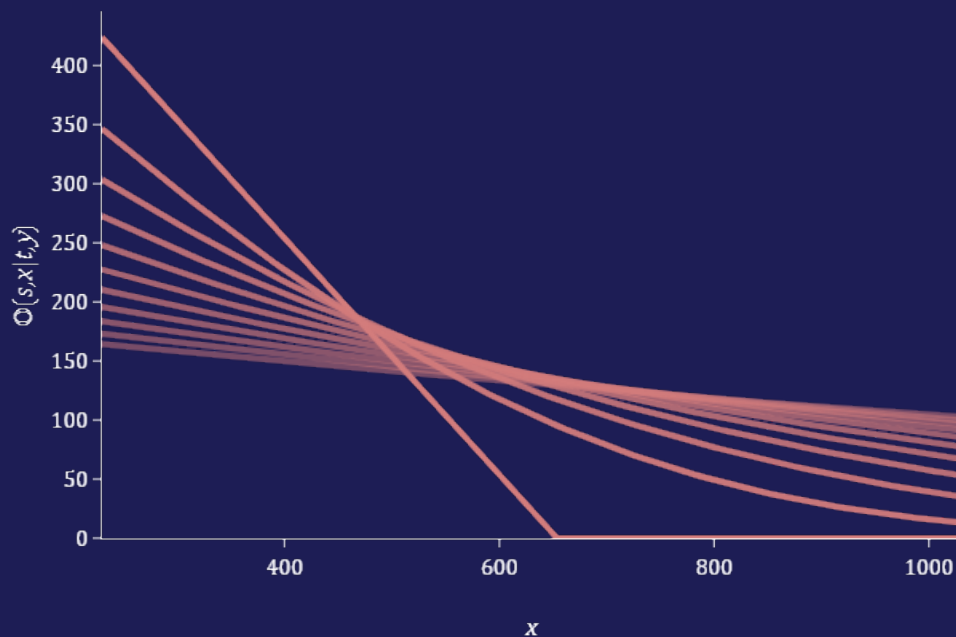


Wheat yields appears to converge quickly, but not immediately. The parameter estimates are in this table.

	rho	mu	sigma	alpha
Parameters	1.54210	867.245	941.732	0.522883

An alpha close to $\frac{1}{2}$ confirms that yields are almost, but not quite stationary. The yield last year will be some, but not much, help in predicting the yield this year.

Multi-peril crop insurance is a Vanilla Put Option. In the plot below, both the horizontal and vertical axes are Wheat in kilograms per hectare. The farmer will receive a payout if yield falls below 654 kilograms per hectare.



The curves on the plot are not flat because Wheat does not converge immediately. For in-the-money insurance on the left, the lowest curve is one year in advance. For out-of-the-money insurance on the right, the highest curve is one year in advance. If the expected yield is 854 kilograms per hectare, insurance will cost 114.5 kilograms per hectare. This is 13.4% of the expected yield. This is the actuarially fair price before the insurer adds their profit margin. Multi-peril crop insurance will be expensive.

See Also: Other Options, Vanilla Put Option

Methods

Methods calculate answers to questions. At the bottom of each method are links to case studies which use the method.

Resilience

Resilience is a wonderful thing. We all strive to be resilient and we all know what it means—until we try to measure it. Perhaps it is the distance to a threshold or tipping point. Perhaps it is resistance to change. Perhaps it is the speed of bouncing back from a shock. Perhaps it is the probability of survival rather than irreversible change.

Or perhaps it is all of these plus something else. The distance to a threshold does not tell us whether we are hurtling toward a cliff, or inching back slowly. Resistance to change may mean we are hanging by our fingernails, or sitting on the porch watching the world go by. Speed tells us nothing about the size of the shock and whether it bounced us off the porch or over a cliff. Probability of survival doesn't say for how long.

There is a measure which summarises these other measures—Passage Times. A Passage Time implies passing over a threshold. The Passage Time may be never or it may be now. If we are hanging by our fingernails it may be soon. If we are sitting on the porch, it may be this afternoon. Rather than the probability of survival, it tells us the probability of how long we might survive.

Not all thresholds are irreversible. We may visit the neighbours and sit on their porch for awhile and then come home. Crossing a reversible threshold is measured by Visiting Time Probabilities. If, however, we are hanging by our fingernails, we may cross an irreversible threshold. Crossing an irreversible threshold is measured by a First Passage Time Probability.

We will summarise Passage Time Probabilities with medians and percentiles. As examples, there is a 50% chance we will return from the neighbours by sundown. There is a 25% chance our fingernails will hold until the helicopter arrives. We may have a few more years to snorkel on the Great Barrier Reef. We may survive long enough to avert global warming. Or not.

Recovery

This tutorial investigates how quickly kangaroo populations recover from shocks.

1. From the 'Ornstein-Uhlenbeck' menu, select the 'Real Options' panel;
2. On the 'Data' tab, from the 'File' list, choose 'Ecosys_Kangaroos';
3. Click on the 'i' button to learn about the data;
4. Close the dialog by clicking the 'Close' button or clicking outside the dialog;
5. From the 'State' list, choose 'Red Pop' and click 'Plot'. Also plot 'Grey Pop' and 'Euro Pop'.

Red Kangaroos are more numerous than Grey Kangaroos are more numerous Euro Kangaroos. All three Kangaroo populations are stochastic.

6. With 'Euro Pop' showing, go to the 'Estimates' tab. Estimation may happen automatically. If not, click 'Go'.

The Euro population converges at rate $\rho = 0.576$ toward a long-term population of $\mu = 479,691$ head. The long-term standard deviation is $\sigma/(2*\rho)^{0.5} = 82,538$ head.

7. This is a bit tricky. Go to the 'Regime' tab. In the lower row of input boxes, set 'phi' to 1;
8. Go to the 'Passage Times' tab. In the lower row of input boxes, set 'omega' to zero.

A positive phi parameter specifies recovery. The omega argument sets the degree of irreversibility. An omega of zero is for a completely reversible threshold.

9. If you are looking at the plot of probabilities, click the '>' button;
10. Click the 'Sync' button.

The Sync button has set threshold k equal to μ and set initial population x to k minus two long-term standard deviations. Starting from two standard deviations below k , there is a 50% chance the Euro population will have reached k within 1.981 years. The other percentiles show times for a 25% chance and a 75% chance.

10. Click the '>' button.

These are percentiles on the plot of the Visiting Time Probability. Notice that the probability does not go to one. It goes to 0.5. Once the population reaches its long-term mean, it will spend 50% of the time above and 50% of the time below the long-term mean. The median is the time when there is a 50% chance the population will be spending 50% of its time above the long-term mean.

11. Click the '_' button to clear any previous sets of arguments and save this set.

Later, we will recall this set of arguments to compare with Red and Grey Kangaroo populations.

12. Go to the 'Estimates' tab, from the 'State' list, choose 'Red Pop'. Click 'Go'.

The Red population converges at rate $\rho = 0.350$ toward a long-term population of $\mu = 1,716,230$ head. The long-term standard deviation is $\sigma/(2*\rho)^{0.5} = 495,387$ head.

13. Go to the 'Passage Times' tab, click the 'Sync' button and click the '≡' button to save this set of arguments with the previous set for the Euro population.

The Red population recovers more slowly than the Euro population.

14. Finally, go once again to the 'Estimates' tab. This time, from the 'State' list, choose 'Grey Pop'. Click 'Go'.

The Grey population converges at rate $\rho = 0.130$ toward a long-term population of $\mu = 914,762$ head. The long-term standard deviation is $\sigma/(2*\rho)^{0.5} = 334,498$ head.

15. Go to the 'Passage Times' tab, click the 'Sync' button and click the '≡' button to save this set of arguments with the previous two sets for the Euro and Red populations;

The Grey population recovers slowest of all.

16. Click the 'V' button or the '^' button and click again to rotate through the results.
17. Click the '<' or '>' button and rotate through the results again.

Euros are more resilient than Reds are more resilient than Greys.

18. In the lower row of input boxes, set 'omega' equal to one. Press 'Enter' or click 'Plot' to recalculate and plot.

Nothing changed. With omega set to one, the threshold is completely irreversible and times are First Passage Times. For the special case of threshold k equal to location μ , the First Passage Time Percentiles equal the Visiting Time Percentiles. Otherwise, they will differ.

These results are consistent with the reproductive biology of kangaroos. When conditions are tough, female Euro and Red Kangaroos can keep a joey in the pouch and pause an embryo. When conditions improve they reproduce quickly and make a fast recovery. Female Grey Kangaroos can't. But all three species are resilient. Other species and systems may be less resilient. They may or may not recover from habitat destruction, drought and climate change.

See Also: Wandering Albatross

Tipping Point

This tutorial estimates the time before Sydney's water storage falls to the volume requiring water restrictions or turning on the desalination plant.

1. From the 'Ornstein-Uhlenbeck' menu, select the 'Real Options' panel;
2. On the 'Data' tab, from the 'File' list, choose 'Ecosys_SydneyWater';
3. Click on the 'i' button to learn about the data;
4. Close the dialog by clicking the 'Close' button or clicking outside the dialog;
5. From the 'Time' list, choose 'Year'. From the 'State' list, choose 'Warragamba'. Press 'Enter' or click 'Plot'.

Warragamba Dam is Sydney's major water storage with a capacity of 2,031 gegalitres, or perhaps a bit more. During droughts, its volume has fallen to below 40% of capacity, prompting water restrictions, and even the construction of a desalination plant.

6. In the input box 'Begin' enter 2025 and press 'Enter' or click 'Plot';
7. Click the 'Reset' button.

You can choose which observations to plot using the 'Begin' and 'End' input boxes. The 'Reset' button plots all observations.

8. Go to the 'Estimates' tab. Estimation may happen automatically. If not, click 'Go'.

The alpha parameter of 0.96 identifies the water volumes in the dam as almost Scaled Brownian Motion with slow convergence toward the long-term mean of 1,792.55 gegalitres, which is about 88% of capacity.

9. Go to the 'Passage Times' tab. In the lower row of input boxes, set 'k' to 894 for gegalitres at 44% capacity, set 'x' to 2031 for initial gegalitres and make sure 'omega' is set to one for First Passage Times. Click 'Plot';
10. If you are looking at a plot of probability, click the '<' or '>' button to look at the other plot;
11. Click 'Axes' to adjust the axes of the plot to fit the results.

The calculations behind this plot are very expensive and the series on the plot may look a bit wobbly. If you are prepared to wait, do step 12 below. Otherwise, skip to step 13.

12. In the upper row of boxes, set 'z:From' to 0 and 'z:To' to 2100. For 'z:By' delete the number and leave the box blank. Click 'Plot' and wait;
13. Click the '_' button to delete previous sets of arguments and save this set;

14. Change the initial water volume, 'x' to 1793 and click 'Plot';
15. Click the '≡' button to also save this set of arguments;
16. Change the initial water volume, 'x' to 1000 and click 'Plot';
17. Click the '≡' button again to save this set of arguments;
18. Click the 'V' button or the '^' button and click again to rotate through the results for initial water volumes.

Even when the dam is full, the median First Passage Time before the water volume reaches 44% capacity is only 6.8 years. When the dam is at its long-term mean, the median time is still 6.3 years. But when the dam is low, the median time is next year.

19. Click the '<' or '>' button and again rotate through the results again.

After experiencing a long drought, WaterNSW considered its options. Although it has been raining lately, the results of this analysis suggests those options may be exercised in a few years.

See Also: Farm Dams and Destocking

Adoption

What's wrong with these people? Why aren't they adopting my amazing new widget? Well, probably nothing and a new widget must compete with existing widgets in any adoption decision.

Researchers take several approaches to studying adoption: Surveys of attitudes and behaviours; Sigmoidal curves from early adopters to laggards; Double hurdles for choosing whether to adopt and, if so, the intensity of adoption; Benefit-Cost analysis; Environmental Benefit-Cost analysis; Cost Effectiveness analysis; Portfolio analysis.

What does a Real Options analysis contribute? First, what doesn't it contribute? It is not a portfolio analysis. It is an actuarially fair evaluation of one technology or management regime which might be included in a portfolio. It is information for making a portfolio decision, not the decision itself. As such, it does not include a decision maker's preferences. Risk averse and risk neutral decision makers will choose different portfolios of actuarially fair options.

Although it is not the decision itself, a Real Options analysis calculates decision thresholds. If a technology or regime is to be included in a portfolio, the decision thresholds specify the optimal adoption and disadoption under uncertainty. Adoption and disadoption are options. A decision maker may or may not exercise them, depending upon whether or not a threshold has been crossed.

Before the name Real Options, it was labelled Investment under Uncertainty. Benefit-Cost analysis can be described as Investment under Certainty. In the parlance of Real Options, a Benefit-Cost analysis defines an Obligation. Once you invest, you may get gains but are obliged to take losses. A Real Options analysis is an Obligation sandwiched between two options. Adoption is an Entry Option. Disadoption is an Exit Option. An Entry Option without an Exit Option leaves the Obligation.

Perhaps people have not adopted your amazing new widget because it doesn't suit their portfolios. Perhaps it suits their portfolios but the decision threshold has

not been crossed. These tutorials calculate decision thresholds under uncertainty and the likely time until the thresholds are crossed.

Entry Option

This tutorial calculates Entry Options, Decision Thresholds and First Passage Times to reach the Decision Thresholds.

Financial options expire. Real Options are perpetual and can be exercised at any time. Choosing time to maximise the option value gives the Option Envelope. Finding the state of nature where the Option Envelope equals the payoff gives the Decision Threshold. At the Decision Threshold, a decision maker is indifferent between holding and exercising the option. Crossing a Decision Threshold is irreversible. The measure of how long before a Decision Threshold is crossed is the First Passage Time.

1. From the 'Ornstein-Uhlenbeck' menu, select the 'Real Options' panel;
2. On the 'Regime' tab, make sure the inputs in the boxes match the numbers in the table below.

	x:From	x:To	x:By		
	-30	30	0.6		
rho	mu	sigma			
0.5	15	15			
y	r	phi	b	c	
0	0.05	1	0	0	

In particular, make sure phi is 1 for an Entry Option. After any changes, press 'Enter' or click 'Plot'.

3. Click on the 'Info' button to learn about Entry Options, Obligations, Prohibitions and Exit Options;
4. At the bottom of the dialog, click 'Close' or click outside the dialog;
5. You may be looking at the Prohibition. Click the '>' button until you see the Entry Option.

The kinked function is the payoff function. The curves are options at times before the payoff. The Entry Option is the envelope of these curves.

6. On the 'Decision Threshold' tab, click the 'Info' button to learn about Decision Thresholds;
7. Click 'Close' to close the dialog;
8. Click the '_' button to clear any previous arguments and save this set of arguments.

The Ornstein-Uhlenbeck Process converges toward the long-term mean of $\mu = 15$. The Decision Threshold is $k = 14.66$ where the Option Envelope equals the payoff. At the Decision Threshold, the Entry Option $\mathbb{O} = 14.66$.

9. In the input box, set 'y' to 5 and press 'Enter' or click 'Plot';
10. Click the '≡' button to save this set of arguments.

With a break even point of $y = 5$, the Option Envelope and the payoff shift right. The Decision Threshold is $k = 16.48$ and Entry Option $\mathbb{O} = 11.48$ which is 5 less than the Decision Threshold.

11. In the input box, set 'b' to 10 and click 'Plot';
12. Click the '≡' button to save this set of arguments.

With a lump sum subsidy of $b = 10$, everything shifts up. The Decision Threshold $k = 15.63$ is a bit less than before. The Entry Option $\mathbb{O} = 20.63$ is equal to $k - y + b$ which is more than before. Mostly, the decision maker pockets the subsidy and makes about the same entry decision.

13. Click the 'V' button to show the original set of arguments;
14. On the 'Passage Times' tab, make sure 'omega' is set to 1 for First Passage Times;
15. Click the 'Sync' button.

The 'Sync' button brings the Decision Threshold into the threshold box, 'k', and sets initial state 'x' to a curious number. This curious number is the Decision Threshold for an Exit Option.

16. Click the '<' button or '>' button to show the plot of percentiles from all possible initial states.

The median is the 0.5 percentile. There is a 50% chance the state will go from exit to entry before a time of 1.42.

17. Return to the 'Decision Threshold' tab. In the input box, change 'sigma' to 5 and click 'Plot'.

Entry is less uncertain and the Decision Threshold is smaller.

18. Back on the 'Passage Time' tab, click 'Sync'.

The median First Passage Time from exit to entry is 2.56. The Ornstein-Uhlenbeck Process is less uncertain, the state is less likely to bounce over the threshold and the median First Passage Time is longer.

The Entry Option values the flexibility to adopt a Regime. Exercising the Entry Option eliminates any Prohibition against adoption.

See Also: Agriculture and Climate, Fisheries and ITQs

Obligation / Prohibition

This tutorial investigates Obligations and Prohibitions.

Real Options are uncertainty sandwiches. Sandwiched between two Options is the Real filling. Between an Entry Option and an Exit Option is an Obligation. Between an Exit Option and an Entry Option is a Prohibition. An Obligation is an investment with real benefits and real costs. A Prohibition is a negative investment, avoiding the costs but foregoing the benefits.

This tutorial repeats the first three steps of the Entry Option tutorial.

1. From the 'Ornstein-Uhlenbeck' menu, select the 'Real Options' panel;
2. On the 'Regime' tab, make sure the inputs in the boxes match the numbers in the table below.

	x:From	x:To	x:By		
	-30	30	0.6		
rho	mu	sigma			
0.5	15	15			
y	r	phi	b	c	
0	0.05	1	0	0	

In particular, make sure phi is 1 for an Entry Option. After any changes, press 'Enter' or click 'Plot'.

3. You may be looking at the Prohibition. Click the '>' button until you see the Entry Option;
4. Click the '_' button to delete any previous arguments and save this set.

The kinked function is the payoff function. The curves are options at times before the payoff.

5. In the input box, set 'phi' to -1 and 'Plot';
6. Click the '≡' button to save this as the second set of arguments.

Again, the kinked function is the payoff function and the curves are options at times before the payoff. Notice that the payoff function for the Entry Option minus the payoff function for the Exit Option is a 45° line.

7. Click either the 'V' button or the '^' button to show the Entry Option;
8. Hover the mouse over a point on a curve at some time before the payoff and record the option price;
9. Click either the 'V' button or the '^' button to show the Exit Option;
10. Hover the mouse over the same 'x' on the corresponding curve;
11. Subtract this option price from the previous option price.

For example, on the second curve at $x = 21$, the Entry Option is 1.998435. The Exit Option is 8.500187. The difference is -6.501752.

11. Click the '<' button or the '>' button to show the Obligation;
12. Hover the mouse over the same 'x' on the corresponding curve.

For the same example, the answer is -6.501752. Repeating this exercise on any curve at any 'x' will get a similar answer. The Obligation is the Entry Option minus the Exit Option.

13. Click the 'V' button or the '^' button to show the Prohibition.
14. Hover the mouse over the same 'x' on the corresponding curve.

This time the answer is a positive 6.501752. Repeating for any x on any curve will show that the Prohibition is the Exit Option minus the Entry Option.

15. In the middle row of input boxes, set 'sigma' to zero and 'Plot';
16. Click the '≡' button to save this as the third set of arguments;
17. Set 'phi' to -1 and 'Plot' to get the Obligation;
18. Click the '≡' to save this as the fourth set of arguments.
19. Click the 'V' or '^' button to rotate through sets of arguments

The Obligation and Prohibition are linear in x and don't change with sigma. Uncertainty does not affect linear functions.

20. Click the '<' or '>' button to view an Entry or Exit Option;
21. Click the 'V' or '^' button to view the effect of 'sigma'.

Both Entry and Exit Options increase with sigma. We are used to subtracting risk premiums because we usually work with concave functions. However, options are everywhere convex in x and we add premiums for risk.

An Obligation is important. It is the real part of a Real Options analysis. But an Obligation is unaffected by uncertainty. Uncertainty is in unreal parts—the Entry and Exit Options.

See Also: Put-Call Parity

Exit Option

This tutorial calculates Exit Options and Decision Thresholds and explores the effects of the Ornstein-Uhlenbeck parameters.

Without an Exit Option, an investor has an Obligation. In olden days, an unlucky investor would end up in debtor's prison. In 1542, the Statute of Bankrupts codified bankruptcy—an Exit Option to encourage investment. These days some investors are so encouraged they declare bankruptcy as often as they can.

1. From the 'Ornstein-Uhlenbeck' menu, select the 'Real Options' panel;
2. On the 'Regime' tab, make sure the inputs in the boxes match the numbers in the table below.

	x:From	x:To	x:By	
	-30	30	0.6	
rho	mu	sigma		
0.5	15	15		
y	r	phi	b	c
0	0.05	-1	0	0

In particular, make sure phi is -1 for an Exit Option. After any changes, press 'Enter' or click 'Plot'.

3. You may be looking at the Obligation. Click the '>' button to see the Exit Option.

The kinked function is the payoff function. The curves are options at times before the payoff. The Exit Option is the envelope of these curves.

4. Go to the 'Decision Threshold' tab. The threshold should calculate automatically. If not, click 'Plot'.

The Decision Threshold is $k = 3.454$ where the Option Envelope equals the payoff. At the Decision Threshold, the Entry Option is $\mathbb{O} = 3.454$.

5. In the middle row of input boxes, set 'sigma' to 0 and press 'Enter' or click 'Plot';
6. Click the '_' button to clear any previous arguments and save this set of arguments;
7. Set 'sigma' to 15, 'Plot' and click the '≡' button to save;
8. Set 'sigma' to 30, 'Plot' and click the '≡' button to save;
9. Click the '^' button to rotate through the arguments and observe the effect of 'sigma'.

With $\sigma = 0$, there is certainty and exit is at $k = 0$. Larger sigma delays the exit.

10. Click the 'V' button to show the second set of arguments;
11. Click the '_' button to erase previous arguments and save this set;
12. Set 'mu' to 0, 'Plot' and click the '≡' button to save;
13. Set 'mu' to -15, 'Plot' and click '≡' to save;
14. Click the '^' button to observe the effect of 'mu'.

If an investment is expected to be profitable, the Exit Option is small.
If is expected to be unprofitable, the Exit Option is large.

15. Click the '^' button to show the first set of arguments;
16. Set 'rho' to 0.1, 'Plot' and click the '_' button to erase and save;
17. Set 'rho' to 0.5, 'Plot' and click '≡' to save;
18. Set 'rho' to 1.0, 'Plot' and click '≡' to save;
19. Click the '^' button to rotate through the results.

If an investment is expected to be profitable, faster convergence reduces the value of the Exit Option.

20. Click the 'V' button or the '^' button to until the first set of arguments appears;
21. Set 'mu' to -15 and repeat the exercise for 'rho' equal to 0.1, 0.5 and 1.0.

If an investment is expected to be unprofitable, faster convergence increases the value of the Exit Option.

The Exit Option values the flexibility to disadopt a Regime. Exercising the Exit Option eliminates any Obligation.

See Also: Agriculture and Climate

Policies

If you have a hammer everything looks like a nail. You build things that can be nailed together. If you are economist, everything looks like marginal revenue or marginal cost. You design policies to equate them.

If everything is concave, equating marginal revenue and marginal cost will maximize benefits above costs and identify the best possible decision. But if everything is convex, equating marginal revenue and marginal cost will maximize costs above benefits and identify the worst possible decision.

Decisions under uncertainty come in small, medium and large. Small decisions are governed by Transition Densities, which are concave where it matters. Medium decisions are governed by Transition Probabilities, which are both convex and concave. Large decisions are governed by the integral of Transition Probabilites, which become Options. Options are everywhere convex in the underlying states of nature.

Small decisions under uncertainty are amenable to marginal analysis using Stochastic Dynamic Programming. Medium decisions can be analysed by comparing probabilities using Stochastic Dominance. Large decisions require Real Options. There are

no marginal analyses or simple comparisons, but global searches for optimal decision thresholds.

Policies for large decisions are not about marginal trade-offs, but about hastening or delaying the crossing of thresholds. If a technology has environmental, cultural or social benefits, policies may hasten its adoption and delay its disadoption. Conversely if a technology has social costs, policies may delay its adoption or hasten its disadoption.

No Exit

No Entry

Taxes and Subsidies

Resource Rent

This tutorial explores how resource rent affects the decision to enter an industry under uncertainty.

If you rent an apartment and decide to move back home, you stop paying rent. If you buy an apartment and decide to move back home, you are obligated to pay fixed costs until you sell it. Fixed costs are what you pay for owning property. Rent is what you pay for using someone else’s property.

1. From the ‘Ornstein-Uhlenbeck’ menu, select the ‘Real Options’ panel;
2. On the ‘Regime’ tab, make sure the inputs in the boxes match the numbers in the table below.

	x:From	x:To	x:By		
	-30	30	0.6		
rho	mu	sigma			
0.5	15	15			
y	r	phi	b	c	
0	0.05	1	0	0	

A positive phi is for an Entry Option. After changes, press ‘Enter’ or click ‘Plot’.

3. Click on the 'Info' button to learn about Entry Options, Obligations, Prohibitions and Exit Options;
4. At the bottom of the dialog, click 'Close' or click outside the dialog;
5. You may be looking at the Entry Option. Click the '<' button or '>' button until you see the Prohibition.

The Prohibition is the benefits of not investing. If the Obligation is positive and the investment is growing, the Prohibition is negative. If the Obligation is negative and the investment is wilting, the Prohibition is positive.

6. In the input box, set interest rate 'r' to 0 and press 'Enter' or click 'Plot'.

The Prohibition is easier to interpret without discounting. At the time an option is exercised, the benefits of not investing are revealed along the 45°. The many other lines are for times further and further from exercising. The Ornstein-Uhlenbeck Process is expected to converge to long-term mean $\mu = 15$. A long time from exercising an option, the benefits of not investing are expected to be -15.

7. Click the '_' button to clear any previous arguments and save this set of arguments;
8. In the input box, set 'y' to 5 and 'Plot';
9. Click the '≡' button to save this set of arguments;
10. In the input box, set 'b' to -10 and 'Plot';
11. Click the '≡' button to save this set of arguments.

Positive break even point y shifts everything to the right. Negative benefits b shifts everything up. For this example, negative b is resource rent. Both the break even point and the resource rent increase the Prohibition— increase the benefits of not entering.

12. Click the 'V' button until the first argument set is showing;
13. In the input box, set interest rate 'r' to 0.05 and 'Plot';
14. Click the '≡' button to save this set of arguments;
15. Again click the 'V' button until the second argument set is showing;
16. In the input box, set interest rate 'r' to 0.05 and 'Plot';
17. Click the '≡' button to save this set of arguments;
18. Finally, click the 'V' button until the third argument set is showing;
19. In the input box, set interest rate 'r' to 0.05 and 'Plot';
20. Click the '≡' button to save this set of arguments;

Discounting reduces the magnitude of benefits and costs. Over a long enough time, discounting reduces everything to zero.

21. On the 'Decision Threshold' tab, click the 'V' button or the 'Λ' button and cycle through the sets of arguments.

On the plots, the Decision Threshold is k and the Entry Option at the threshold is $\hat{O}$. The break even point and resource rent change k very little but decrease $\hat{O}$ substantially. Comparing argument set 1 with argument set 4, discounting shifts k to the left. Curiously, it advances rather than delays entry. Comparing argument set 3 with argument set 6, discounting has little effect because there is not much to discount.

Companies will enter the industry and pay resource rent, as long as the rent is set properly. Properly means allowing for the costs of doing business and uncertainty and leaving the Entry Option positive. Industry would prefer no rent, of course. Resource owners would prefer the largest rent that still allows entry. Resource owners don't know industry's costs and can't just invent a rent. This is why the truth telling mechanism of auctions is important.

On a different subject—since the climate started warming, the economics profession has been in a crisis of confidence. Discounting a benefit/cost analysis of global warming discounts catastrophe into 'no worries mate, she'll be right'. In a Real Options analysis, discounting is a factor, but uncertainty and thresholds dominate.

See Also: Fisheries and ITQs

Insurance

Finance

Finance is all about contracts between willing buyers and willing sellers. Real Options have no contracts. A financial option expires. A real option is perpetual. A financial option is governed by calendar time. A real option allows the choices of if and when.

But the same calculations can be used for financial options, just with a fixed interpretation of time.

Vanilla Call Option

This tutorial compares option pricing with the Ornstein-Uhlenbeck Process to option pricing with Brownian Motion. It uses a European call option as its example.

In finance, Geometric Brownian Motion is the most commonly used stochastic process. It does not converge. Perhaps it applies to stock prices. But commodity prices don't sail away. They bob about but are anchored by supply and demand.

1. From the 'Ornstein-Uhlenbeck' menu, select the 'Analytical' panel;
2. On the 'Option' tab, make sure the inputs in the boxes match the numbers in the table below.

		x:From	x:To	x:By	s:By
		-30	30	0.6	0.1
	rho	mu	sigma	t	s:To
	0	15	15	10	10
y	r	phi	b	~	s:From
0	0.05	1	0	0	1

If you make any changes, press 'Enter' or click 'Plot'.

3. In the middle row of input boxes, make sure 'rho' is set to 0. In the bottom row of boxes, make sure 'phi' is set to 1. Press 'Enter' or click 'Plot'.

With $\rho = 0$, this is Scaled Brownian Motion, a special case of Arithmetic Brownian Motion. For $\phi = 1$, this is a call option.

4. Click the '<' button or the '>' four times to scroll through the available plots.

You can hover the mouse over plots to see the values. You can hold the mouse down on a 3D plot and twirl it around to view from different angles. You can zoom with the middle mouse button. And there are more controls at the upper right.

5. Click the '_' button to delete any previous arguments and save these arguments;
6. In the middle row, set 'mu' to 0 and click 'Plot'.

The plot didn't change with mu. If the rate of convergence, rho, equals zero, the location, mu, disappears from all calculations, leaving only scale, sigma.

7. Click the 'V' button to undo the change to 'mu'. Set 'rho' to 0.5 and 'Plot'.

In this case, the Ornstein-Uhlenbeck Process converges toward location mu equal to 15. Compared with Scaled Brownian Motion, the option is much more likely to expire in the money, even if the initial price is out of the money. An out-of-the-money option is more expensive. Conversely, a deeply in-the-money option is less valuable.

8. Click '≡' to also save these arguments;
9. Click the '<' button or the '>' button to view your favourite plot;
10. Click the 'V' button or the '^' button to compare Scaled Brownian Motion to a convergent Ornstein-Uhlenbeck Process;

11. Hover the mouse in the upper right corner of your favourite plot. Click on the left-most icon, the camera icon.

You have downloaded the plot to your default download folder. If the Moon is glowing in the menu, the plot will be for a dark background. If the Sun is shining, the plot will be for a light background.

12. Set 'rho' to 99 and click 'Plot'.

In this case, the Ornstein-Uhlenbeck Process converges immediately and loses all connection to underlying commodity price x . Even just before expiry, and regardless of the terminal value, the option price is 15. At earlier times, it is less than 15 due to discounting.

Because the Ornstein-Uhlenbeck Process converges, options written on commodity prices will flat line as the time to expiry increases. The option price becomes independent of the current commodity price.

See Also: Carbon Contracts

Vanilla Put Option

This tutorial demonstrates hedging with options on futures, after estimating the Ornstein-Uhlenbeck Process using real data.

Commodity prices may converge toward long-term prices determined by demand and supply. Futures prices may converge as well. Options on commodity futures should account for this convergence.

1. From the 'Ornstein-Uhlenbeck' menu, select the 'Maximum Likelihood' panel;
2. On the 'Estimates' tab, from the 'File' list, choose 'Finance_KansasCity_WheatFutures';
3. Click the 'i' button to read about the data;
4. At the bottom of the dialog click 'Close' or click outside the dialog;
5. From the 'State' list, choose 'Close' for closing prices and press 'Enter' or click 'Go'.

These are the unrestricted estimates for wheat futures as an Ornstein-Uhlenbeck Process. In the table, a positive rho is required for convergence. However, the size of rho does not directly indicate the speed of convergence. Its value depends upon how time is measured. The alpha parameter is determined by both rho and time. It indicates the speed of convergence. An alpha equal to 1 identifies Scaled Brownian Motion which does not converge; an alpha equal to $\frac{1}{2}$ indicates a stationary process which has converged every time it is observed. Wheat futures are in between but are closer to Scaled Brownian Motion.

6. Click the 'Reset' button;

The plot shows more data. By default, the plot shows 200 observations. The 'Reset' button shows all observations. The 'Begin' and 'End' input boxes will show a subset of the data. The estimates do not change, however. Estimates always use all observations.

7. Back through the 'Ornstein-Uhlenbeck' menu, select the 'Analytical' panel;
8. On the 'Option' tab, in the lower row of input boxes, set 'y' to 540 for the strike price, 'r' to 0.0002 for the daily interest rate and 'phi' to -1 for a put option. If you know brokerage costs you can enter them under 'c';
9. In the upper row, set 'x:From' to 500 and 'x:To' to 600. For 'x:By', delete the number in the box and leave it blank;
10. In the far right column, set 's:From' to 0 and 's:To' to 60. For 's:By', delete the number in the box;
11. In the middle row, set 't' to 60.;
12. Click 'Plot' to redraw. Click the '<' button or the '>' button to view 3D plots. On the 3D plot, itself, click the mouse and drag to rotate.

The plots show the option prices for all possible wheat futures at all times starting from 60 days before expiry. If you wish to offset the option before expiry, 's:To' can stop before 't'.

13. Click the '_' button to delete any previous arguments and save these arguments;
14. Back through the 'Ornstein-Uhlenbeck' menu, again select the 'Maximum Likelihood' panel;
15. On the 'Likelihood Ratio Test' tab, in the bottom row, set 'rhor' to 0 and click 'Go'.

These are restricted estimates for wheat futures as Scaled Brownian Motion.

16. Back through the 'Ornstein-Uhlenbeck' menu, select the 'Analytical' panel;
17. You should be on the 'Option' tab looking at a plot of option prices for Scaled Brownian Motion. If not, click 'Plot';
18. Click '≡' to also save these arguments. Click '∧' and click again to compare option prices for the Ornstein-Uhlenbeck Process and Scaled Brownian Motion.

With convergence, options in the money are worth much less. Wheat futures converge toward location μ above strike price y . At 60 days from expiry, options in the money are worth about the same as options out of the money.

Even if convergence is slow, option pricing for the Ornstein-Uhlenbeck Process is very different from option pricing with Brownian Motion.

Put-Call Parity

This tutorial shows how put and call options can be transformed from one to the other and back.

Who makes up this jargon. Put-Call Parity means there is a linear solution in between put and call options. Adding the linear solution transforms a put into a call. Subtracting transforms a call into a put. Because the solution is linear in the underlying price, it does not protect from upside or downside risks. All risk protection is in the options.

This tutorial continues with the wheat futures example of the previous tutorial. The first 11 steps below may be repetitive.

1. From the 'Ornstein-Uhlenbeck' menu, select the 'Maximum Likelihood' panel;
2. On the 'Estimates' tab, from the 'File' list, choose 'Finance_KansasCity_WheatFutures';
3. Click the 'i' button to read about the data;
4. At the bottom of the dialog click 'Close' or click outside the dialog;
5. From the 'State' list, choose 'Close' for closing prices and press 'Enter' or click 'Go';
6. Back through the 'Ornstein-Uhlenbeck' menu, select the 'Analytical' panel;
7. On the 'Option' tab, in the lower row of input boxes, set 'y' to 540 for the strike price, 'r' to 0.0002 for the daily interest rate and 'phi' to -1 for a put option. If you know brokerage costs you can enter them under 'c';
8. In the upper row, set 'x:From' to 500 and 'x:To' to 600. For 'x:By', delete the number in the box and leave it blank;
9. In the far right column, set 's:From' to 0 and 's:To' to 60. For 's:By', delete the number in the box;
10. In the middle row, set 't' to 60;
11. Click 'Plot' to redraw.

This replicates the previous tutorial for a put option with a strike price of 540.

12. Click the '<' button until the 2D plot is in view;
13. On the 2D plot, hover the mouse over a point on a curve and make a note of option price ① for that point.

For example, a point on the second curve for x equal to 520 has an ① equal to 13.21571.

14. In the lower row of boxes, set 'phi' to 1 for a call option;
15. Make a note of option price ② for the same x on the corresponding curve;
16. Subtract the smallest option price from the largest.

For x equal to 520 on the second curve, ② equals 2.946476. The difference is 10.26924.

17. On the 'Obligation / Prohibition' tab, hover over the plot for the same x on the corresponding curve and find this same difference;
18. In the lower row of boxes, set ' ϕ ' to -1 for a put option;
19. For the same x on the corresponding curve, find the difference on this plot.

The Obligation has the same difference but negative. You can repeat this exercise for any x on any of the curves.

The difference is the Put-Call Parity. A call option minus a put option is linear in the underlying price. Obviously, the payoff function for a call option minus the payoff function for a put option is a 45° line. But any curve on the 2D plot for a call option minus the corresponding curve on the plot for a put option is a straight line. The angle becomes less than 45° at times before expiry, due to convergence at rate ρ and discounting at rate r . If both rates are set to zero, all lines are 45°.

In finance, the Put-Call Parity is a curiosity. In Real Options it is crucial. It is the actual investment and is called the Obligation. The options either side of the Obligation help manage the risks.

Strangle and Straddle

This tutorial calculates strangles and straddles to protect against price movements.

Buying a call option and selling a put option at the same strike price is the Put-Call Parity. No one would actually do such a thing. There is no hedge against price movements, only brokerage costs.

Buying a call option and buying a put option at the same strike price is a Strangle. It protects against all price movements, both up and down. Buying a call option at a higher strike price and buying a put option at a lower strike price is a Straddle. It protects against more extreme price movements.

This tutorial continues with the wheat futures example of previous tutorials. The first 11 steps below may be repetitive.

1. From the 'Ornstein-Uhlenbeck' menu, select the 'Maximum Likelihood' panel;
2. On the 'Estimates' tab, from the 'File' list, choose 'Finance_KansasCity_WheatFutures';
3. Click the 'i' button to read about the data;
4. At the bottom of the dialog click 'Close' or click outside the dialog;
5. From the 'State' list, choose 'Close' for closing prices and press 'Enter' or click 'Go';
6. Back through the 'Ornstein-Uhlenbeck' menu, select the 'Analytical' panel;
7. On the 'Option' tab, in the lower row of input boxes, set ' y ' to 540 for the strike price, ' r ' to 0.0002 for the daily interest rate and ' ϕ ' to -1 for a put option. If you know brokerage costs you can enter them under ' c ';
8. In the upper row, set ' x :From' to 500 and ' x :To' to 600. For ' x :By', delete the number in the box and leave it blank;
9. In the far right column, set ' s :From' to 0 and ' s :To' to 60. For ' s :By', delete the number in the box;
10. In the middle row, set ' t ' to 60;
11. Click 'Plot' to redraw.
12. Click the '<' button until the 2D plot is in view;

13. On the 2D plot, hover the mouse over a point on a curve and make a note of option price $\mathbb{O}$ for that point.
14. In the lower row of boxes, set 'phi' to 1 for a call option;
15. Record option price $\mathbb{O}$ for the same x on the corresponding curve;

For x equal to 520 on the second curve, $\mathbb{O}$ for a put option equals 13.21571 and $\mathbb{O}$ for a call option equals 2.946476. In the previous tutorial, we subtracted. This time we will add to get 16.16219.

16. From the 'Ornstein-Uhlenbeck' menu, select the 'Finite Difference' panel;
17. On the 'Option' tab, click on the 'Info' button to learn about the arguments for calculating option prices using the Finite Difference method;
18. At the bottom of the dialog, click 'Close'.
19. From the 'Formula' list, make sure 'Kinked' is chosen;
20. In the bottom row of input boxes, set 'xo' to 540 for the kink point and 'Vmax' to 100. Press 'Enter' or click 'Plot';
21. On the 2D plot, hover the mouse over the same x on the corresponding curve as before and make a note of option price $\mathbb{O}$.

For x equal to 520 on the second curve, $\mathbb{O}$ equals 13.21836. This number can vary depending upon the computer, but is close to 13.21571 from Analytical option pricing. The Finite Difference method is less accurate. Three or four significant digits is about what can be expected.

22. In the lower row of input boxes, set 'vs' to 1 for a call option and click 'Plot';
23. On the 2D plot, make a note of $\mathbb{O}$ at the same x on the corresponding curve;
24. Add the recorded option prices.

For x equal to 520 on the second curve, $\mathbb{O}$ equals 2.949125 for a call option. Adding $\mathbb{O}$ for a put option gives 16.167485 which is very close to the previous answer.

25. From the 'Formula' list at the top, choose 'Butterfly';
26. In the bottom row of boxes, set 'xo' to 540, 'xm' to 540 and 'Vmax' to 100. Click 'Plot';
27. On the 2D plot, hover the mouse over the same x on the corresponding curve as before and make a note of option price $\mathbb{O}$.

For x equal to 520 on the second curve, $\mathbb{O}$ equals 16.16749 for a straddle which is nearly the same as calculating the call and put options separately and adding.

28. In the bottom row of input boxes, set 'xm' to 560 for the strike price of the call option. Click 'Plot'.

This is a straddle. It is less valuable than a strangle because it affords less protection.

For strangles and straddles, call and put options can be calculated separately by Analytical formulas and added up. Or the sum can be calculated in one go using the Finite Difference method.

Bear Spread

This tutorial calculates a bear spread to protect against price falls.

A bear spread requires shooting a bear and tanning the hide. Or buying a put option at a higher strike price and selling a put option at a lower strike price to offset the cost. It cushions moderate falls.

The option price of a bear spread equals the price of the purchased option minus the price of the sold option. Or, it can be calculated using the Finite Difference method.

1. From the 'Ornstein-Uhlenbeck' menu, select the 'Maximum Likelihood' panel;
2. On the 'Estimates' tab, from the 'File' list, choose 'Finance_Commodities';
3. Click the 'i' button to read about the data;
4. At the bottom of the dialog click 'Close' or click outside the dialog;
5. From the 'Time' list choose 'Year'. From the 'State' list, choose 'WTI' and press 'Enter' or click 'Go'.

These are the unrestricted estimates for WTI as an Ornstein-Uhlenbeck Process. The alpha parameter of 0.95 identifies this as a slowly converging process, but not quite Scaled Brownian Motion.

6. Back through the 'Ornstein-Uhlenbeck' menu, select the 'Finite Difference' panel;
7. On the 'Option' tab, from the 'Formula' list, make sure 'Kinked' is chosen;
8. In the lower row of input boxes, set 'xo' to 70 for the higher strike price, 'vs' to -1 for put options, 'Vmax' to 20 for a lower strike price of 50 and 'Vmin' to 0. If you know brokerage costs you can enter them under 'VMin' as a negative number;
9. In the upper row, set 'x:From' to 0 and 'x:To' to 150. For 'x:By', delete the number in the box and leave it blank;
10. In the far right column, set 's:From' to 2025 and 's:To' to 2026. For 's:By', delete the number in the box;
11. Click 'Plot' to redraw.

These are the option prices for a bear spread with underlying WTI prices converging to location, μ , above the strike price.

12. Click the '_' button to replace existing arguments with these arguments;

13. From the 'Ornstein-Uhlenbeck' menu, select the 'Maximum Likelihood' panel;
14. On the 'Likelihood Ratio Test' tab, in the lower row of input boxes, set 'rho' to 0 and click 'Go';
15. Back through the 'Ornstein-Uhlenbeck' menu, select the 'Finite Difference' panel;
16. You should be viewing the 'Option' tab and the option prices should have recalculated. If not, click 'Plot';

These are the option prices for a bear spread if the underlying WTI prices are Brownian Motion.

17. Click the '≡' button to save these arguments. Compare with the previous result by clicking the '^' button.

With convergence toward location μ , which is above the strike price, in-the-money options below the strike price are less valuable.

For a bear spread, like other options, even slow convergence has a big effect on option prices.

See Also: Weather Derivative

Weather Derivative

This tutorial demonstrates option pricing if the underlying stochastic process is biophysical, not financial.

A weather derivative is a bear spread with weather as the underlying stochastic process. If weather is observed at infrequent intervals it becomes approximately stationary. Its price will differ from the price of an option written on Arithmetic or Geometric Brownian Motion.

1. From the 'Ornstein-Uhlenbeck' menu, select the 'Maximum Likelihood' panel;
2. On the 'Estimates' tab, from the 'File' list, choose 'Agric_SA_GMErroroo';
3. Click the 'i' button to read about the data;
4. At the bottom of the dialog click 'Close' or click outside the dialog;
5. From the 'State' list choose 'Apr.Oct.rain' and press 'Enter' or click 'Go'.

These are the unrestricted estimates for growing season rainfall as an Ornstein-Uhlenbeck Process. The alpha parameter close to $\frac{1}{2}$ identifies this as a rapidly converging process which is almost stationary.

6. Back through the 'Ornstein-Uhlenbeck' menu, select the 'Finite Difference' panel;
7. On the 'Option' tab, from the 'Formula' list, make sure 'Kinked' is chosen;
8. In the lower row of input boxes, set 'xo' to 250 for millimetres of growing season rainfall required for a crop of wheat, 'vs' to -0.02 for the damage per millimetre below 250, 'Vmax' to 1.5 tonnes of wheat per hectare and 'Vmin' to 0;

9. In the upper row, set 'x:From' to 0 and 'x:To' to 500. For 'x:By', delete the number in the box and leave it blank;
10. In the far right column, set 's:From' to 2025 and 's:To' to 2026. For 's:By', delete the number in the box;
11. Click 'Plot' to calculate and redraw;
12. Using the '<' button or the '>' button, scroll through the 2D and 3D plots.

Suppose a weather derivative is purchased in October 2025 for expiry in October 2026. The growing season rainfall for 2025 has been observed. The rainfall system converges quickly, but not immediately and the rainfall this year informs the pricing of the weather derivative for next year. Suppose it was an extraordinarily wet year with 400 millimetres. The actuarially fair price is 0.5472 tonnes per hectare, about 30% of the maximum yield of 1.5 tonnes per hectare.

13. In the lower row of boxes, set x_0 to '200' for a lower strike price and click 'Plot'.

As expected the weather derivative is less valuable. The actuarially fair price with October 2025 growing season rainfall of 400 millimetres is 0.2344 tonnes per hectare.

14. Click the '_' button to delete any previous arguments and save these arguments;
15. From the 'Ornstein-Uhlenbeck' menu, select the 'Maximum Likelihood' panel;
16. On the 'Likelihood Ratio Test' tab, in the lower row of boxes, set ' ρ ' to 0 and click 'Go';
17. Back through the 'Ornstein-Uhlenbeck' menu, select the 'Finite Difference' panel;
18. You should be viewing the 'Option' tab and the option prices should have been recalculated. If not, click 'Plot'.

These are the actuarially fair prices for growing season rainfall as Scaled Brownian Motion. For 400 millimetres of October 2025 growing season rainfall, this is 0.01345 tonnes per hectare.

20. Click the '≡' button to also save these arguments. Click the '^' button to compare with the previous results.

With a strongly convergent process, there is a tenuous connection to the past. In-the-money options below the strike price are less valuable and out-of-the-money options far above the strike price are more valuable.

Option pricing, even in finance, does not require the underlying stochastic process to be measured in dollars. Any biophysical process can be estimated and its option price calculated. The option price can be in dollars or not. In this tutorial, the weather derivative is in tonnes of wheat per millimetre of growing season rainfall. The price

of wheat could be hedged separately by a put option and exchange rates could be hedged by a put or call option. Or this could all be rolled together by a broker and sold to the farmer as one contract.

See Also: Bear Spread

Other Options

This tutorial demonstrates the Finite Difference method for option prices with any payoff function. The example is a yield index which is kinked in two places and non-linear elsewhere.

1. From the 'Ornstein-Uhlenbeck' menu, select the 'Maximum Likelihood' panel;
2. On the 'Estimates' tab, from the 'File' list, choose 'Agric_SA_GMClare';
3. Click the 'i' button to read about the data;
4. At the bottom of the dialog click 'Close' or click outside the dialog;
5. From the 'State' list choose 'Apr.Oct.rain' and press 'Enter' or click 'Go'.

These are the unrestricted estimates for growing season rainfall as an Ornstein-Uhlenbeck Process. The alpha parameter close to $\frac{1}{2}$ identifies this as a rapidly converging process which is almost stationary.

6. Back through the 'Ornstein-Uhlenbeck' menu, select the 'Finite Difference' panel;
7. On the 'Terminal Values' tab, from the 'Formula' list, choose 'Transcendental';
8. In the lower row of input boxes, set 'xo' to 100 for the lowest possible millimetres of growing season rainfall, 'xi' to 250 for the inflection point, 'xm' to 450 for the millimetres where yields are maximised, 'Vmax' to 5 tonnes of wheat per hectare and 'Vmin' to 0;
9. In the upper row, set 'x:From' to 0 and 'x:To' to 800. For 'x:By', delete the number in the box and leave it blank;
10. Click 'Plot' to calculate and redraw.

Too little and too much growing season rainfall damage wheat yields. 450 millimetres is just right. The shape and height of the curve can be adjusted for each paddock.

11. Still on the 'Terminal Values' tab, from the 'Formula' list, choose 'Yield Index';
12. Repeat the arguments for this terminal value, 'xo' to 100, 'xi' to 250, 'xm' to 450, 'Vmax' to 5 and 'Vmin' to 0.
13. Click 'Plot' to calculate and redraw.

This is the Transcendental function upside down. It could be the payoff function for yield index insurance, but farmers are unlikely to insure against all possible losses.

14. In the lower row of boxes, set 'Vmax' to 4 and 'Vmin' to -1;
15. Click 'Plot' to calculate and redraw.

The payoff will be capped at 4 tonnes per hectare.

16. On the 'Option' tab, in the far right column, set 's:From' to 0 and 's:To' to 1. For 's:By', delete the number in the box;
17. Press 'Enter' or click 'Plot';
18. Click the '<' button or the '>' button to view 2D and 3D plots. Click and drag the mouse to rotate 3D plots. Zoom with the middle mouse button.

Growing season rainfall converges quickly toward location mu of 484.9 millimetres. This is in the sweet spot. If in October of this year, growing season rainfall was 200 millimetres, yield index insurance for the next growing season will cost 0.3212 tonnes per hectare, or 6% of the maximum yield of 5 tonnes per hectare. If growing season rainfall was 704 millimetres, yield index insurance for the next growing season will cost 0.2806 tonnes per hectare, or 5% to 6% of the maximum yield.

As with weather derivatives, a fixed price could be attached to the yield index, or price risk could be hedged separately.

See Also: Yield Index Insurance

Contracts to Deliver

This tutorial calculates the price of the uncertainty underlying a contract.

Financial options are contracts between willing buyers and sellers, but are continuously marked to market. Either party can offset their position and exit at any time. The value of a financial option is its Option Price.

Other contracts are not marked to market and cannot be offset. For example, producers may enter a contract to deliver goods at a fixed price. In the parlance of Real Options, they have an Entry Option but no Exit Option. The lack of an Exit Option may make producers reluctant to enter into contracts and make the contracts less valuable.

1. From the 'Ornstein-Uhlenbeck' menu, select the 'Maximum Likelihood' panel;
2. On the 'Estimates' tab, from the 'File' list, choose 'Agric_SA_WaiteRotationTrial';
3. Click the 'i' button to read about the data;
4. At the bottom of the dialog click 'Close' or click outside the dialog;
5. From the 'State' list choose 'Plt 11-16 WWPPPP P' and press 'Enter' or click 'Go'.

These are kilograms per hectare of Pasture in a Wheat/Pasture rotation. The alpha parameter close to $\frac{1}{2}$ identifies this series as almost, but not quite, stationary. The mu parameter is the long-term mean. The long-term standard deviation is $\sigma/(2\rho)^{\frac{1}{2}} = 2595$. This is about 31% of the long-term mean. Pasture production is relatively risky.

6. Back through the 'Ornstein-Uhlenbeck' menu, select the 'Analytical' panel;
7. On the 'Mean' tab, in the lower row of input boxes, set 'x' to 8717.63. In input boxes to the right, set 't:From' to 0 and 't:To' to 1. For 't:By', delete the number in the box and leave it blank;
8. Press the 'Enter' key or click 'Plot';
9. On the plot, hover the mouse over the Mean at time 1.

Pasture production this year of 8717.63 kilograms per hectare is expected to decline next year to 8400 kilograms per hectare. With this expectation, a farmer is considering an agistment contract.

10. On the 'Decision Threshold' tab, set 'y' to 5522 and 'phi' to 1. Click 'Axes' to adjust the axes, calculate and plot.
11. Click the ' _ ' button to clear any previous arguments and save this set.

Conveniently, one Dry-sheep equivalent (DSE) will consume 1 kilogram of pasture per day. The horizontal axis is both kilograms per hectare and days of agistment. Doubly conveniently, the price for agistment per day is about \$1.00 per DSE. The vertical axis is all three, kilograms per hectare, days of agistment and dollars. The lucky number entered for y gives a Decision Threshold, k, equal to the expected production. The Option Price at the Decision Threshold, $\hat{O}$, equals $k - y$, or \$2878. A farmer will enter a contract for 5522 days of agistment—if they can also exit the contract.

12. Set 'phi' to -1 for an Exit Option and press 'Enter' or click 'Plot'.

Given the option, a farmer will offset the contract if pasture production falls to $k = 4940$ kilograms per hectare and pay a penalty of $\hat{O} = \$582.1$.

13. Click the 'V' button to recall the previous arguments;
14. Set 'y' to 5490, set 'phi' to -1 and 'Plot';
15. Hover the mouse over the Option Envelope at 8400.

A farmer whose pasture is expected to produce 8400 kilograms per hectare, will value this Exit Option at \$165.6.

16. Set 'phi' to 1 and 'Plot';
17. The input box for 'b' should now be active. Set 'b' to -165.6 and 'Plot' again;
18. Click the '≡' button to save this set of arguments;

Setting $b = -165.6$ subtracts the value of the Exit Option from the value of the Entry Option at the expected pasture production.

19. Click the 'V' button or the '^' button to compare results.

A farmer who expects pasture production to be 8400 kilograms per hectare and has an Exit Option will contract for 5522 days of agistment. A farmer who has no Exit Option will contract for 5490 days of agistment. A farmer who has an Exit Option will value the flexibility to enter into an agistment contract at \$2878. A farmer who has no Exit Option will value the flexibility at \$2744.

A pastoralist who wants to agist sheep on a farmer's pasture and wants the farmer to deliver and not default, must pay a bit extra and accept a bit less. In this example, the pastoralist must pay \$134 more for 32 fewer days of agistment. Otherwise, a farmer will not accept the contract. Conversely, a pastoralist could take the risk of default and pay \$134 less for 32 more days of agistment.

See Also: Carbon Contracts

Time to Stop Loss

This tutorial demonstrates how to anticipate the time before hitting a stop loss order.

Traders don't seem to worry about time, but they could. A stop loss order is a threshold which liquidates a position in the market. This threshold might be considered irreversible, which suggests calculating First Passage Times.

Passage Times are the flip side of a coin toss. On one side, prices are stochastic and time is fixed. On the other side, time is stochastic and prices are fixed. There is no certain time to reach a stop loss order. There are First Passage Time Probabilities and Percentiles.

1. From the 'Ornstein-Uhlenbeck' menu, select the 'Real Options' panel;
2. On the 'Estimates' tab, from the 'File' list, choose 'Finance_KansasCity_WheatFutures';
3. Click the 'i' button to read about the data;
4. At the bottom of the dialog click 'Close' or click outside the dialog;
5. From the 'State' list choose 'Close' and press 'Enter' or click 'Go'.
6. On the 'Passage Times' tab, in the lower row of input boxes, set 'k' to 525 for the stop loss, 'x' to 575 for the current wheat futures price and 'omega' is 1 for First Passage Times;
7. Click 'Axis' to rescale the axis, recalculate and plot;
8. Click the '<' or '>' button to view the other plot.

There is a 25% chance the futures price will hit the stop loss before 21.73 days, a 50% chance it will hit before 35.40 days and a 75% chance it will hit before 52.72 days.

8. Set 'k' to 625 and click 'Plot'.

There is a 25% chance the futures price will hit this stop loss before 36.25 days, a 50% chance it will hit before 50.48 days and a 75% chance it will hit before 67.64 days. The times are greater in this case because wheat futures converge away from stop loss k toward location μ .

9. Click the 'Info' button to read about 'Passage Times';
10. Instead of clicking the 'Close' button, at the bottom, click the 'More' button.

This takes you to the 'Analytical' panel and the 'Passage Time Percentiles' tab which has more arguments to debate with and plots to look at. The 3D plots don't look like much until you click on the legend.

11. If you have some spare time, in the upper row of input boxes, for 'z:By', delete the number in the box and 'Plot';

The plots are less chunky.

Anticipating the time to a threshold would seem to be useful for planning ahead. Not only for planning trades, but for planning life.

Estimation

Estimation of the Ornstein-Uhlenbeck Process is unlike other statistical methods. If you are a guru in time-series analysis, you will find these tutorials to be frustrating. If you are a novice, you will find them to be easy.

Other statistical methods do not estimate the rate of convergence. Time-series methods, such as ARIMA and GARCH, assume stationarity to eliminate the rate of convergence and estimate only the location and scale. Location is often confused with the Mean and scale squared is often confused with the Variance.

Relaxing the assumption of stationarity changes everything. The rate of convergence must be estimated. Location is in the Means and scale is in the Variances, but the rate of convergence is in both. There isn't just a Mean and a Variance. There is a unique Mean and a unique Variance for every observation. The rate of convergence links Means and Variances and determines the probability distribution of the estimates.

So don't try to squeeze these tutorials into your knowledge of time-series analysis. They won't fit.

Rate of Convergence

This tutorial demonstrates the estimation of the rate of convergence, ρ , and its relationships with the location parameter, μ , and the scale parameter, σ .

The rate of convergence determines the behavior of the Ornstein-Uhlenbeck Process. If it converges slowly, the Ornstein-Uhlenbeck Process is like a random walk; the current state of nature is the best predictor of future states. If the Ornstein-Uhlenbeck Process converges rapidly, it is like an ergodic or stationary process; the connection to the current state of nature is lost and the constant location is the best predictor of future states.

1. From the 'Ornstein-Uhlenbeck' menu, select the 'Maximum Likelihood' panel;
2. On the 'Data' tab, from the 'File' list, choose 'OUP_Convergence';
3. Click on the 'i' button to learn about the data;
4. Close the dialog by clicking the 'Close' button or clicking outside the dialog;
5. From the 'State' list, choose different time-series and 'Plot' them.

The time-series are Monte-Carlo simulations with three rates of convergence, ρ , of 0.1, 0.5 and 2.5. As the rate of convergence increases, the series become more jagged.

6. On the 'Log Likelihood' tab, pretend to be a computer. Enter numbers in the input boxes to estimate ' ρ ', ' μ ' and ' σ ';

The goal is to maximize the Log Likelihood, LnL . Because LnL is negative, the goal is to move it closer to zero.

7. Click on the 'Estimates' tab to check your work.

The true location, μ , is -15 for all series. The true scale, σ , has been adjusted so that each series has the same asymptotic variance of 225. In other words, $\sigma = 15 \cdot (2 \cdot \rho)$. The plot shows the state, z , and the Mean, G . G is always between z and location μ . For slow convergence, it is near z . For fast convergence, it is near μ . Otherwise, it is somewhere in between.

8. On the plot legend, click on z and G to hide them;
9. Click on H^2 to show it.

Most observations are over the same length of time and have the same Variance. Some observations are over longer times and have larger Variances. Slow rates of convergence have the smallest Variances. Fast rates of convergence have the largest. The estimated Variance can exceed the true asymptotic Variance of 225.

10. On the plot legend, click on H^2 to hide it;
11. Still on the plot legend, click on v to show it.

Series v are the Standardized Errors. Larger errors over longer intervals are divided by larger Variances. For all rates of convergence, v is between approximately -2 and +2.

12. Re-estimate for the data series with ρ of 0.1, 0.5 and 2.5, but this time observe parameter α .

Parameter α is determined by the rate of convergence, ρ and the average length of an observation interval. Parameter α varies between 0.5 and 1. When it is 0.5, the series is stationary or ergodic. When it is 1, the series is Brownian Motion, or a random walk. When it is in between, the series converges. The smaller is α the faster is the convergence.

The rate of convergence and the length of an observation interval determine the parameter α . Parameter α summarises whether the series is stationary or a random walk or something in between. It is also the shape parameter for the probability distribution of the Log Likelihood. The tutorial on Hypothesis Tests has more.
See Also: Hypothesis Tests

Observation Interval

This tutorial interprets the rate of convergence, ρ , and the scale, σ , depending upon the observation interval.

It is impossible to tell if ρ and σ are large or small—if the Ornstein-Uhlenbeck Process converges quickly or slowly, is erratic or predictable—without knowing how time is measured.

1. From the 'Ornstein-Uhlenbeck' menu, select 'Maximum Likelihood';
2. On the 'Data' tab, from the 'File' list, choose 'OUP_ObservationInterval';
3. Click on the 'i' button to learn about the data;
4. Close the dialog by clicking the 'Close' button or clicking outside the dialog;
5. From the 'Time' list, choose 'Day' and click 'Plot'. Choose 'Year' and click 'Plot'.

The only change in the plot of the data is the units on the horizontal axis.

6. Click on the 'Estimates' tab. Estimate for 'Days' and for 'Years';

Rate of convergence, rho, and scale, sigma, are smaller and larger, depending upon Days or Years. Nothing else changes. Both sets of estimates have the same Log Likelihood, LnL, and the same shape parameter, alpha, which determines the probability distribution of the Log Likelihood.

It may be easier to interpret a plot with years on the horizontal axis. A rho of 0.5, however, is easier to interpret than a rho of 182.5. You can do all the estimation and analysis with time in days but switch to years to draw the plot.

No Missing Observations

This tutorial demonstrates the effect of filling in for observations that aren't missing.

Estimation of the Ornstein-Uhlenbeck Process does not assume weak stationarity and does not require equal time intervals. There are no missing observations—just observations over different lengths of time.

1. From the 'Ornstein-Uhlenbeck' menu, select 'Maximum Likelihood';
2. On the 'Estimates' tab, from the 'File' list, choose 'OUP_NotMissing';
3. Click on the 'i' button to learn about the data;
4. Close the dialog by clicking the 'Close' button or clicking outside the dialog;
5. Press 'Enter' or click 'Go'.

These are the estimates of rho, mu and sigma for observation intervals of equal length.

6. From the 'State' list, choose 'Unequal' and click 'Go'.

The only noticeable difference is a smaller Log Likelihood, LnL, because there are fewer observations. The alpha parameters are almost the same, which means the Log Likelihoods from the 'Equal' and 'Unequal' observations have almost the same probability distributions.

7. On the plot, click on the legend to hide z and G and show H^2 .

The Variance is larger for observations over longer time intervals. How much larger depends upon the length of the observation interval.

8. From the 'State' list, choose 'Average' and click 'Go'
9. Also from the 'State' list, choose 'Means' and click 'Go'.

The series called 'Average' fills in for the non-missing observations with the average of the states of nature. The series called 'Means' fills in with Means, G, as the maximum likelihood estimates of future states of nature. Both sets of estimates seem reasonable.

10. On the plot, compare the Variances, H^2 , for the four series, 'Equal', 'Unequal', 'Average' and 'Means'.

The Variances for Average and Means are smaller. These data are smoothed.

Filling in for observations that aren't missing reduces the estimated Variances and may bias your statistical results. But mostly it is waste of time.

Smoothed Data

This tutorial demonstrates the effect of smoothed data on the parameter estimates of the Ornstein-Uhlenbeck Process.

Physical scientists believe God doesn't play dice with the universe and uncertainty is an error in the model. The easiest way to improve the model is to smooth the data.

1. From the 'Ornstein-Uhlenbeck' menu, select 'Maximum Likelihood';
2. On the 'Estimates' tab, from the 'File' list, choose 'OUP_SmoothedData';
3. Click on the 'i' button to learn about the data;
4. Close the dialog by clicking the 'Close' button or clicking outside the dialog;
5. Press 'Enter' or click 'Go'.

The series 'Data' is simulated with true $\rho = 0.5$, $\mu = -15$ and $\sigma = 15$. For this exercise we will pretend it is raw data.

6. From the 'State' list, choose 'G.uno' and press 'Enter' or click 'Go'. Choose 'G.dos' and click 'Go', 'G.tres' and click 'Go' and so on. Observe the plot as you iterate through the series.

The most noticeable change in the plot is a squeezing of the vertical axis.

7. From the 'State' list, again iterate through 'G.uno', 'G.dos', 'G.tres' and so on. This time observe the table with parameters ρ , μ and σ and the log likelihood LnL .

Parameters ρ and μ change little, but σ goes toward zero and LnL becomes positive.

A small σ is a tell-tale sign the data have been smoothed. A positive LnL is a sure sign. A positive LnL means the Likelihood, itself, is greater than 1. In other words, there is a greater than 100% chance of observing the data as a random sample.

For decisions under uncertainty, smoothed data gives false confidence. Use raw data always. If always isn't possible, estimate ρ and μ and replace the estimated σ with your best guess.

Sample Size

This tutorial demonstrates the effect of sample size on discovering the true parameters of the Ornstein-Uhlenbeck Process.

Data can be hard to find. Large samples are better, of course, but serviceable parameter estimates can be found with fewer observations.

1. From the 'Ornstein-Uhlenbeck' menu, select 'Maximum Likelihood';
2. On the 'Estimates' tab, from the 'File' list, select 'OUP_SampleSize';
3. Click on the 'i' button to learn about the data;
4. Close the dialog by clicking the 'Close' button or clicking outside the dialog;
5. Press 'Enter' or click 'Go' and note the estimates of rho, mu and sigma in the table.

The data are simulated with true $\rho = 0.5$, $\mu = 15$ and $\sigma = 50$.

6. From the 'State' list, choose 'Medium' and click 'Go'. Observe the estimates of rho, mu and sigma;
7. Also from the 'State' list, choose 'Large' and click 'Go'. Observe the estimates of rho, mu and sigma;

The parameters approach the true parameters as the sample size increases.

Experimental scientists have a rule for the minimum sample size. Here, we don't have a rule. The estimation algorithm requires at least three observations for the three parameters, rho, mu and sigma. Of course, larger is better. Meaningful hypothesis tests require reasonable sample sizes. You might check out the tutorial Hypothesis Tests and use the 'OUP_SampleSize' data to test hypotheses.

See Also: Hypothesis Tests

Hypothesis Tests

This tutorial tests hypotheses about parameters.

Hypothesis tests for the Ornstein-Uhlenbeck Process are very different from the usual χ^2 tests.

They are Gamma tests, with χ^2 and Erlang tests as special cases. Ergodic and stationary processes are Invariant and have a χ^2 distribution. Scaled Brownian Motion has an Erlang distribution. The Ornstein-Uhlenbeck Process can be either or neither. A Goodness-of-Fit test compares the unrestricted estimates to the two special cases. A Likelihood Ratio test compares the unrestricted estimates to less restrictive cases.

1. From the 'Ornstein-Uhlenbeck' menu, select the 'Maximum Likelihood' panel;
2. On the 'Goodness-of-Fit' tab, click the 'Info' button to read a summary;
3. Close the dialog by clicking the 'Close' button or clicking outside the dialog;

4. From the 'File' list, choose 'OUP_Convergence';
5. Click the 'i' button to learn about the data;
6. Close this dialog;
7. Press 'Enter' or click 'Go' and look at the lower table of results.

Goodness-of-Fit tests two null hypotheses. R^2 is greater if a null hypothesis is rejected. $1-P$ is smaller if a null hypothesis is rejected. For this data, we can reject the null hypothesis that the Ornstein-Uhlenbeck Process is Invariant at the 8.2% level. We cannot reject the null hypothesis that the Ornstein-Uhlenbeck Process is Scaled Brownian Motion.

8. From the 'State' list, choose other series and test the 'Goodness-of-Fit'.

For this data, the null hypothesis that the Ornstein-Uhlenbeck Process is Scaled Brownian Motion is never rejected.

9. Iterate through the series again, but this time watch the alpha parameter in the upper table of results and compare it with the rho parameter.

Parameter $\alpha = \frac{1}{2}(1 + \exp(-\rho(t-s)))$, where $t-s$ is the length of an observation interval. If ρ is zero, α is 1. If ρ goes to infinity, α approaches $\frac{1}{2}$. Parameter α identifies the probability distribution of -2LnL , where LnL is the Log Likelihood. An α of $\frac{1}{2}$ identifies a Chi^2 distribution. An α of 1 identifies an Erlang distribution. These are special cases of a Gamma distribution with an α from $\frac{1}{2}$ to 1.

10. From the 'File' list, choose 'Agric_NSW_SoilHealthHarden';
11. Press the 'i' button to learn about the data;
12. Close this dialog;
13. From the 'State' list, iterate through soil nitrogen, 'N' and soil water, 'W', under two tillage scenarios, 'Burn' and 'Bash', and test the 'Goodness-of-Fit' by clicking 'Go'.

These series are experimental results collected over long observation intervals and are nearly Invariant. Take a peek at the plot on the Log Likelihood tab to see what that means.

14. Again from the 'Goodness-of-Fit' tab, from the 'File' list, choose 'Finance_Commodities';
15. Click the 'i' button to learn about the data;
16. Close this dialog;
17. Iterate through the series on the 'State' list, clicking 'Go' to test the 'Goodness-of-Fit' for metals and oil commodities.

These series are prices and are almost Scaled Brownian Motion. Take a peek at the plot on the Log Likelihood tab to see what that means.

18. On the 'Likelihood Ratio Test' tab, click the 'Info' button to read a summary;
19. Close this dialog;
20. In the row of input boxes, set 'rhor' to 0, make sure 'mur' and 'sigmar' are blank and click 'Go';
21. Compare the 'Likelihood Ratio Test' with the 'Goodness-of-Fit';
22. On the 'Likelihood Ratio Test' tab, set 'rhor' equal to 99999 and click 'Go';
23. Compare this 'Likelihood Ratio Test' with the 'Goodness-of-Fit'.

Goodness-of-Fit is two Likelihood Ratio tests for rho equal to zero and rho approaching infinity.

24. On the 'Likelihood Ratio Test' tab, change the value of 'rhor' by trial and error to find the 95% upper bound on rho.

For the series WTI, the 95% upper bound on rho is approximately 0.0680.

25. Click 'Reset'. Set 'rhor' to 0 and 'mur' to 99999 and click 'Go'. Set 'rhor' to 0 and 'mur' to 0 and click 'Go'.

If rho is zero, location, mu, cancels from all calculations and can be set to anything. Only if rho cannot be set to zero, can hypothesis tests be conducted on mu.

26. click 'Reset'. By trial and error, set 'sigmar' to find the 95% upper bound for sigma. Find the 95% lower bound for sigma.

For WTI, the 90% confidence interval for sigma is approximately 0.533 to 7.63.

27. From the 'File' list, revisit 'Agric_NSW_SoilHealthHarden';
28. Set 'sigmar' to 99999 and click 'Go'.

OK so this is getting weird. If rho can go to infinity, so can sigma. The asymptotic variance is $\sigma^2/(2*\rho)$, which remains constant as both rho and sigma go to infinity. In other words, estimating a stationary process does not estimate σ^2 , but rather the asymptotic variance. We usually call σ^2 the variance, but it almost never is. Only if $\rho = 0.5$ and the Ornstein-Uhlenbeck Process is observed over long time intervals does the variance converge to σ^2 .

29. Set 'sigmar' to -99999 (notice the minus sign) and click 'Go'.

Parameter sigma can be positive or negative because σ^2 is always positive.

30. By trial and error, find a lower bound for sigma.

For the tillage treatment N.Burn the 95% lower bound is approximately ± 79.1 .

The usual χ^2 tests reject all null hypotheses for medium to large sample sizes. They are overly sensitive. Tests for the Ornstein-Uhlenbeck Process reject far fewer hypotheses. Indeed, they are probably not sensitive enough. You might have to be more flexible than just adopting the 5% rule.

If you are publishing a journal article, reviewers won't question a χ^2 test. All the information is on the Likelihood Ratio Test tab. The test statistic would be $2*(\text{LnLu} - \text{LnLr})$ and the degrees of freedom would be $k_u - k_r$. The critical values are in your favourite table.

Why not just report χ^2 tests? For one thing, degrees of freedom don't exist for tests about rho and sigma and may only exist for tests about mu if observation intervals are of equal lengths. For another, the test statistics have Gamma distributions. If you are keen, the Reference on the Help menu has a long, unreadable appendix called Hypothesis Tests. This is an active area of development and things may change.

My Data

This tutorial shows how to upload your own data. It's easy. Collecting data is the hard part.

Data are read from a Comma Separated Values (.csv) file. There are a few things to watch out for. Excel routinely puts in phantom columns and rows. If this happens, you have to delete the phantoms and save the file again. If you are baffled, open the file in Notepad. There are also 3 different ways to save a worksheet as a .csv file. Hopefully, any of the three will work.

By default, the first column is time and the second column is states of nature. You can put these in different columns, but must remember to select a time column and a state column. The data can be messy with sporadic times and blanks, even text, in odd places. The data will be cleaned and sorted by time. But some of your precious data may be discarded.

1. From the 'Ornstein-Uhlenbek' menu, select the 'Maximum Likelihood' panel;
2. On the 'Data' tab, to the far right, click on the 'Info' button to read about data;
3. If you haven't already, create a .csv file;
4. Still on the 'Data' tab, below the 'File' list, click on the '...' button and browse to your file. 'Open' it;

In the table to the right, the 'First' and 'Last' times will be shown and the number of 'Rows' and 'Cols' will be counted. Don't worry if 'First' is bigger than 'Last'. Data will be sorted.

5. If the times and counts are OK, click on the 'Plot' button. If the plot looks OK, continue. If not, edit the file and repeat from the previous step.

To keep the plot readable, only the last 200 observations are plotted by default. You can change this in the Begin and End boxes. The Reset button will show all the data.

6. On the 'Estimates' tab, click on 'Plot' if necessary to plot the estimates with the data.

A dimmed plot needs to be refreshed. As with all plots, you can click on the legend to show and hide series. You can download by hovering in the upper right to show the icons. The left-most icon is the camera icon for downloads. The background of the plots is transparent. If you are pasting against a dark background, make sure your browser is in dark mode with the Moon glowing in the menu. If you are pasting against a light background, make sure your browser is in light mode with the Sun shining in the menu.

7. On the 'Goodness-of-Fit' tab, check the fit of the Ornstein-Uhlenbeck Process;
8. On the 'Likelihood Ratio Test' tab, test any other interesting hypotheses by entering restricted values.

Hypothesis tests for the Ornstein-Uhlenbeck Process are peculiar. You may want to check out the tutorial, Hypothesis Tests.

Your estimates will follow you wherever you go. The unrestricted estimates will follow you from the 'Estimates' and 'Goodness-of-Fit' tabs. The restricted estimates will follow you from the 'Likelihood Ratio Test' tab. For example, if you want to view a plot of the restricted estimates, go to the 'Log Likelihood' tab after the 'Likelihood Ratio Test' tab. If you want to calculate option prices, from the 'Ornstein-Uhlenbeck' menu, select the 'Analytical' panel and go to the 'Option' tab.

See Also: Hypothesis Tests
